

Complex spatial association rules

Method checks, atlas alignment, and automatically selected findings

GI acute GVHD atlas · colon and duodenum · 3–4 distinct cell types

Reading key. $A \rightarrow B$ means a conditional spatial association, not cell motion or causality. Attraction has Lift above 1; avoidance has Lift below 1. The 598,093-cell dataset contains 267 fields of view (FOVs), 73 biopsies, and 71 patients after exclusion. Unless stated otherwise, an eligible FOV contains at least 20 cells of each named type. Stored occurrences pass mining FDR ≤ 0.05 ; investigation gates are support ≥ 0.01 and confidence ≥ 0.5 for attraction, or expected support $P(A)P(B) \geq 0.01$ for avoidance. Rules with the same arrow sides are pooled across mining centers by averaging their metrics within each FOV. Reversing the arrow defines a different rule. Cell-percentage panels check eligibility only.

Evidence key. Counts are rule-bearing/eligible FOVs; patient counts are distinct people, not independent fields. Reported q values are Benjamini–Hochberg-adjusted patient-level stage or Lift tests from the indicated screen. None of the new stage/strength contrasts below reaches $q < 0.05$; they are exploratory. The parent-reversal screen is descriptive and has no q value. Mining FDR printed inside some maps is an occurrence filter, not a cohort-level test.

1 Method proof

1.1 Longer rules add measured information

Some attractions survive comparison with their shorter parts.

How stored longer-rule occurrences are classified

Three/four distinct cell types · passing investigation gates · one count per FOV
attracts

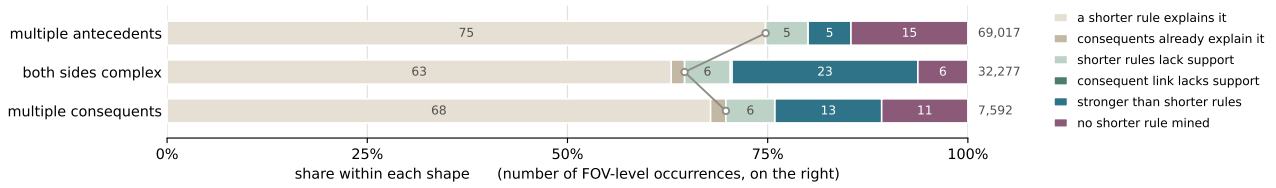


Figure 1: Stored center-specific FOV occurrences are assigned to mutually exclusive information classes by comparing the complex rule with measured shorter rules in that FOV. Counts are occurrences, not unique rules or patients; colours identify added-information classes and grey identifies explained occurrences.

Avoidance also contains rules that add conditional structure

How stored longer-rule occurrences are classified

Three/four distinct cell types · passing investigation gates · one count per FOV
avoids

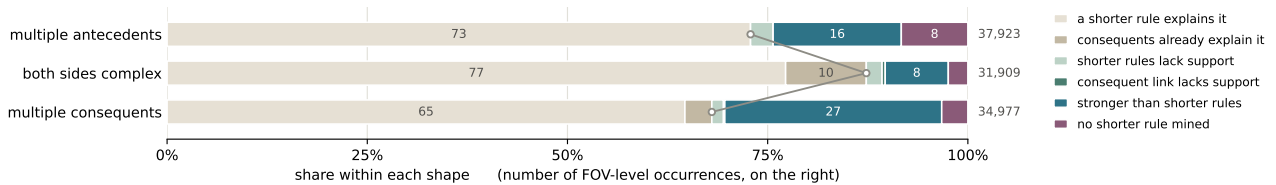


Figure 2: Same occurrence-level classification as Figure 1; lower Lift means stronger avoidance.

A four-type colon attraction exceeds a measured shorter attraction in nine matched fields.

Colon · Endothelial + Epithelial → Goblet + SMV

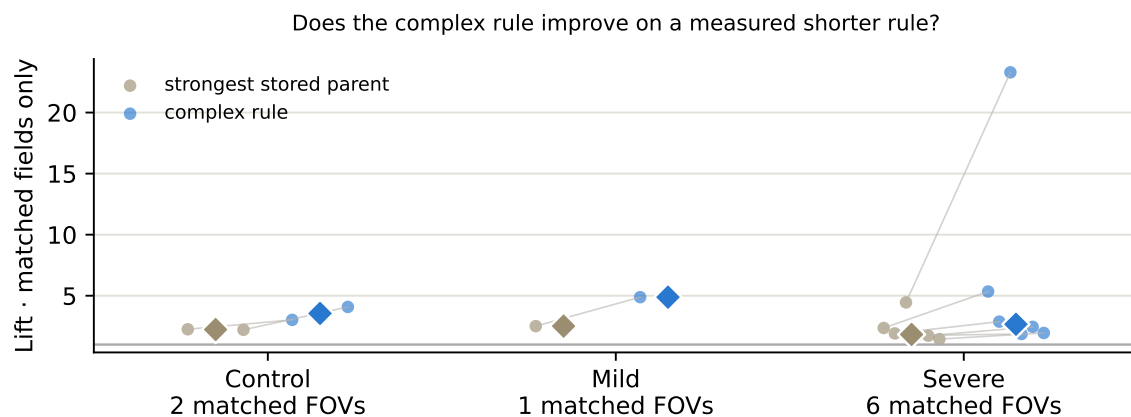


Figure 3: Endothelial+Epithelial → Goblet+SMV. Each joined pair is one FOV's complex Lift and strongest measured same-state shorter-rule Lift (maximum for attraction, minimum for avoidance); diamonds are stage medians. There are 2 control, 1 mild, and 6 severe matched fields. Missing parents have no imputed Lift.

The attraction is visible in the full colon field and its counted cell neighbourhoods.

Colon · Endothelial + Epithelial → Goblet + SMV

Attraction examples · Severe

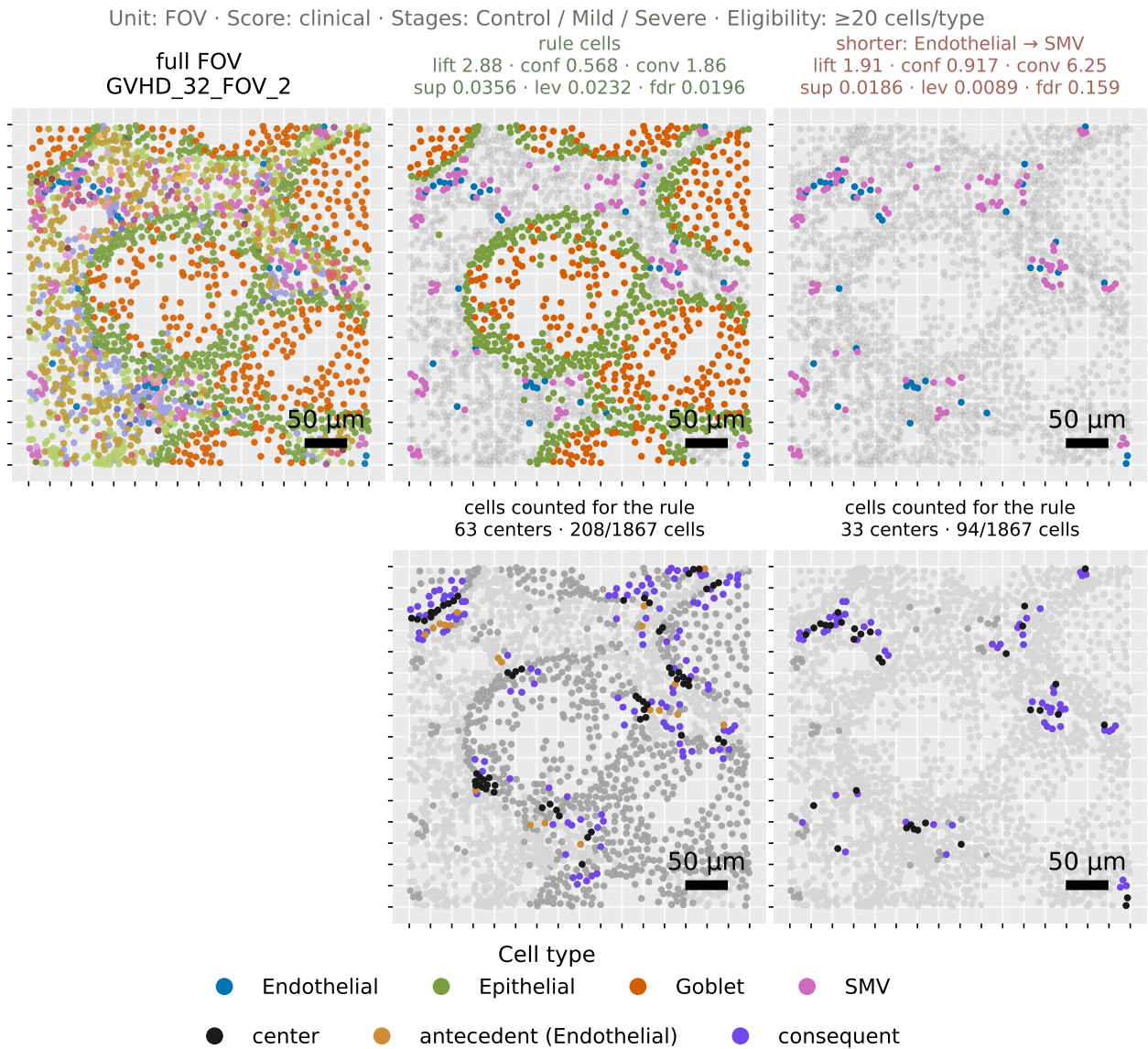


Figure 4: First tissue-map example. Each row is one automatically chosen representative FOV near its stage's median rule Lift. Columns show the full FOV, named rule cell types, and the cells counted for the rule; the latter highlights rule centres and neighbours, not direct cell contact. Text above each row gives the field ID, occurrence metrics and filter status.

Adding plasma context strengthens a measured colon macrophage–muscle avoidance.

Colon · Macrophage + Plasma → Muscle

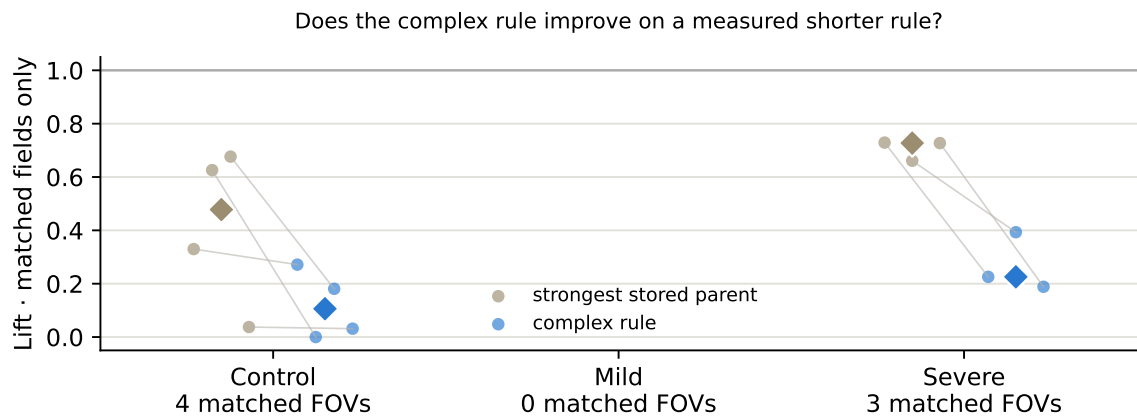


Figure 5: Macrophage+Plasma → Muscle. Same-FOV parent comparison as Figure 3; seven matched fields make this a method demonstration.

The corresponding avoidance can be inspected across the whole tissue field.

Colon · Macrophage + Plasma → Muscle

Avoidance examples · Control

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

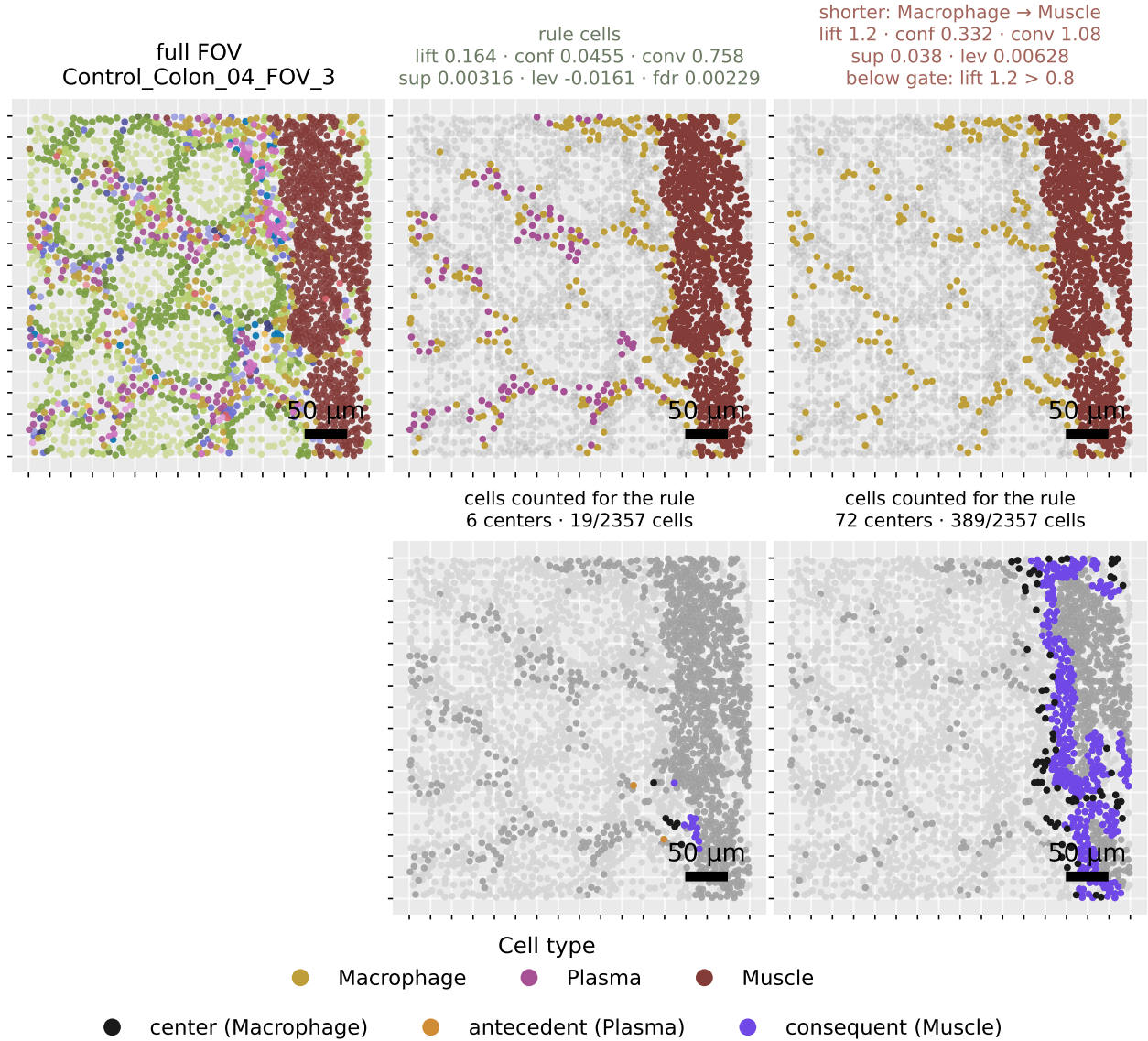


Figure 6: Map layout and selection follow Figure 4.

1.2 A known crypt arrangement

Paneth cells and mesenchymal cells support the intestinal crypt niche (Farin et al., 2016) (Degirmenci et al., 2018). This literature-selected check is retrospective; the broad epithelial and fibroblast labels cannot identify stem cells or specific stromal subtypes.

Paneth attraction to epithelium and fibroblasts appears in 22 of 81 eligible duodenal fields.

Duodenum · Paneth → Epithelial + Fibroblast

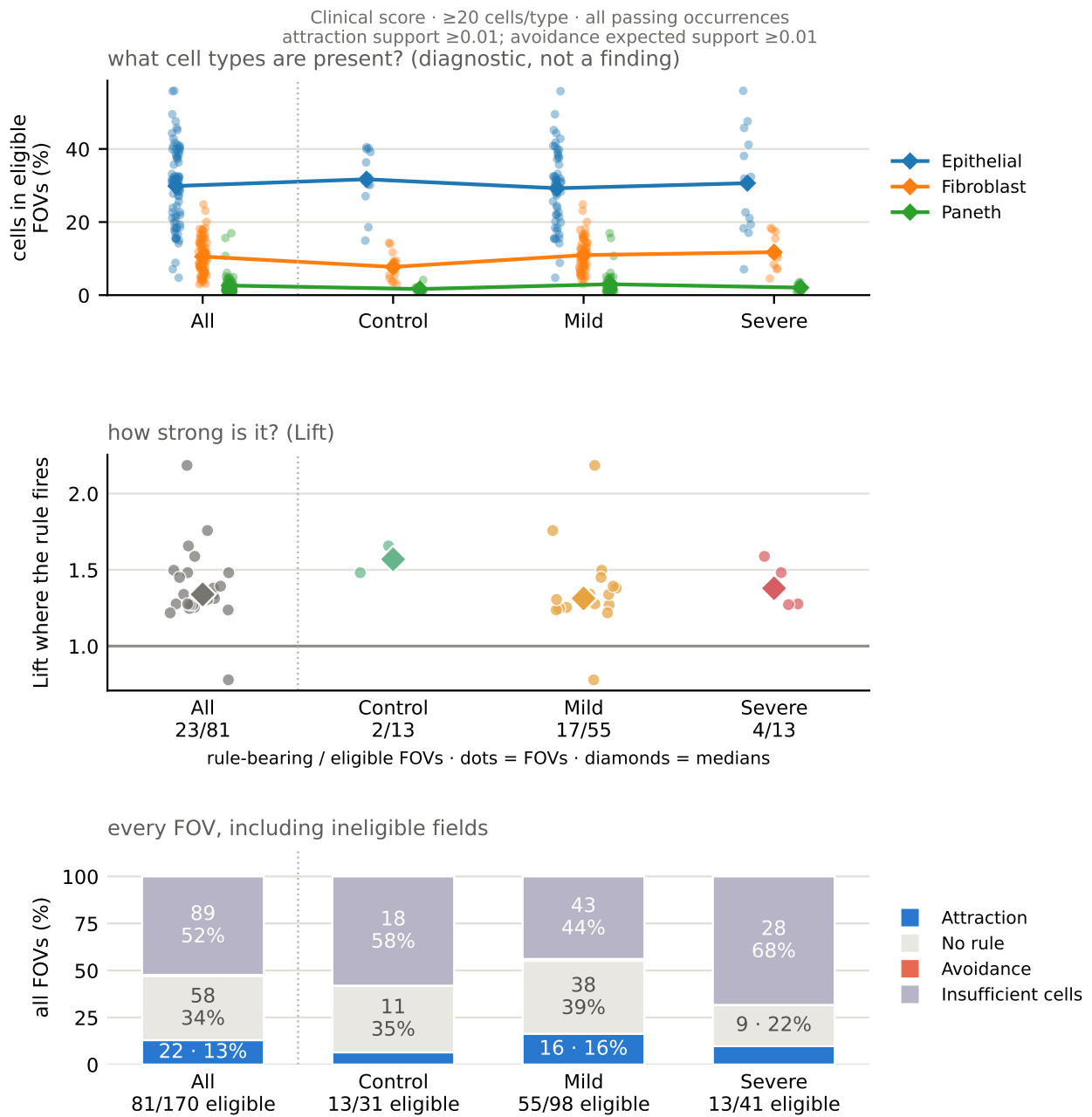


Figure 7: First rule summary. Top: each cell type's percentage within eligible FOVs, with dots for fields and lines/diamonds for stage medians; this is an eligibility diagnostic. Middle: Lift dots for rule-bearing fields and diamond medians, with rule-bearing/eligible counts. Bottom: every FOV partitioned into attraction, avoidance, eligible with no passing rule, or insufficient cells. This known-niche view includes all passing occurrences; only 7/81 are informative beyond shorter rules.

The shorter Paneth-to-epithelium attraction appears in 44 of the same 81 fields.

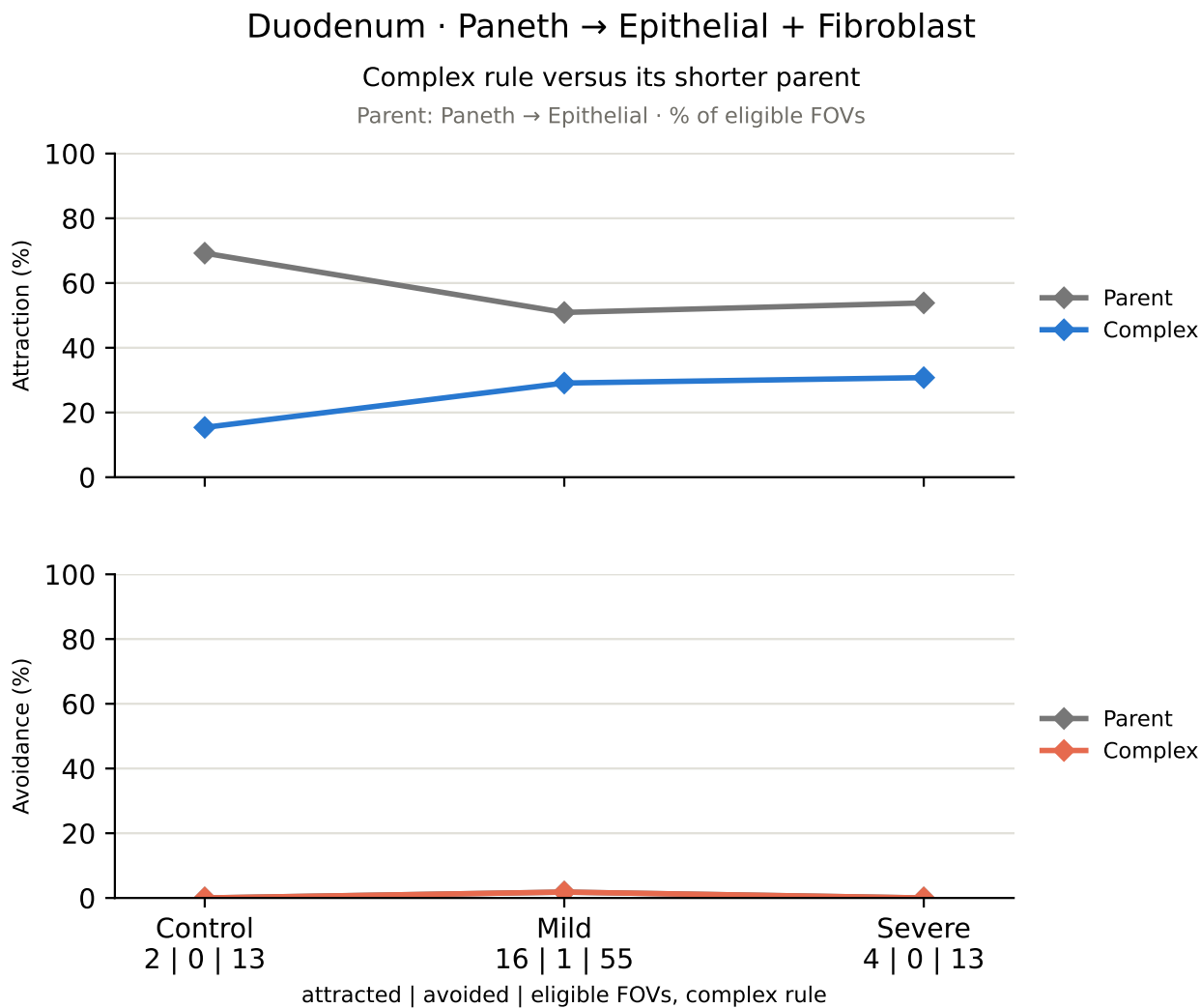


Figure 8: First parent-prevalence comparison. Both measured rules use the complex rule’s eligible FOVs; each point is passing-rule FOVs divided by eligible FOVs at that stage. The shorter rule is fixed before comparing stages.

The crypt rule is visible in intact duodenal fields.

Duodenum · Paneth → Epithelial + Fibroblast

Attraction examples · Control

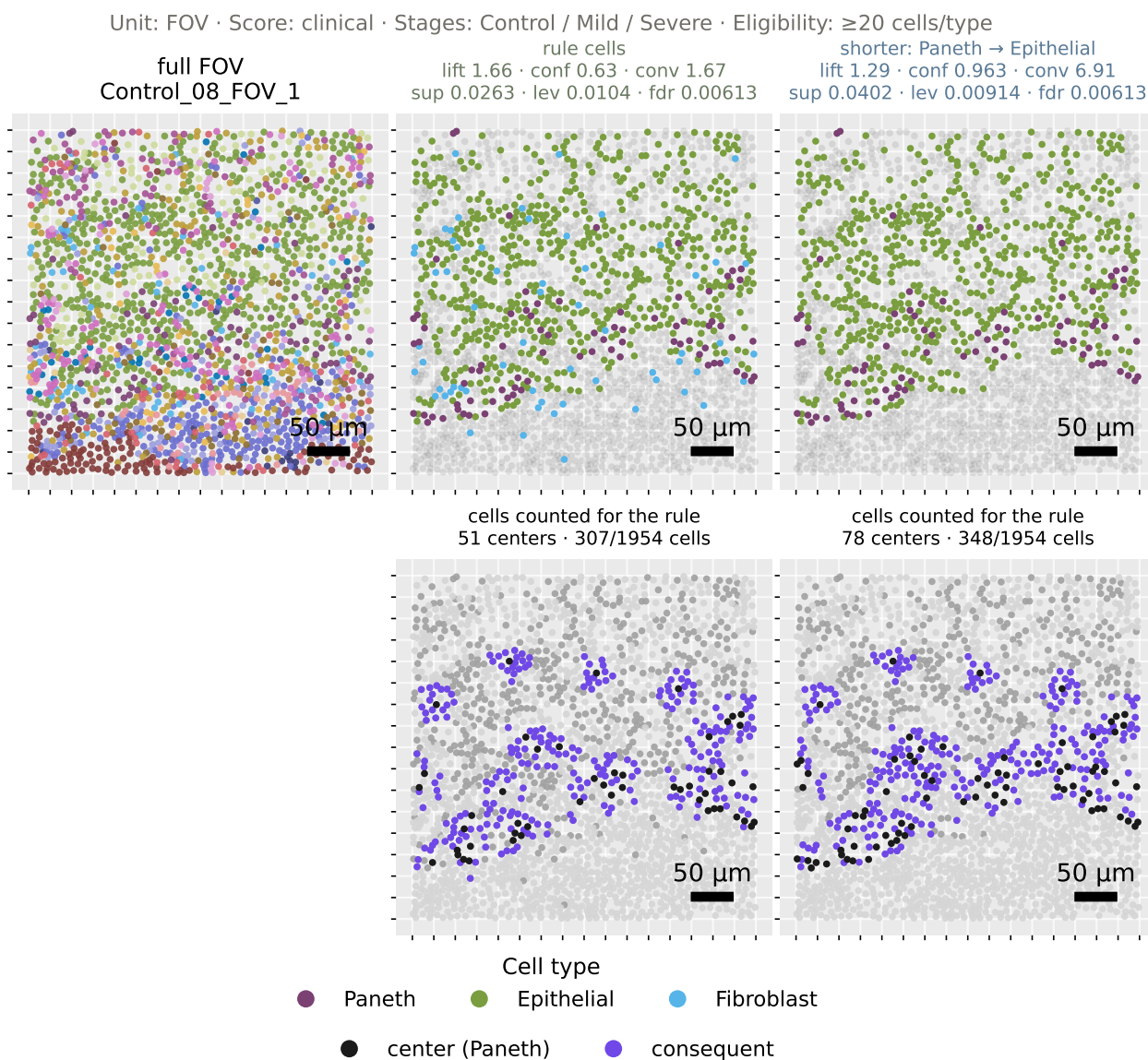


Figure 9: Representative map as in Figure 4.

Adding Paneth context gives one common orientation of this niche.

Duodenum · Fibroblast + Paneth → Epithelial

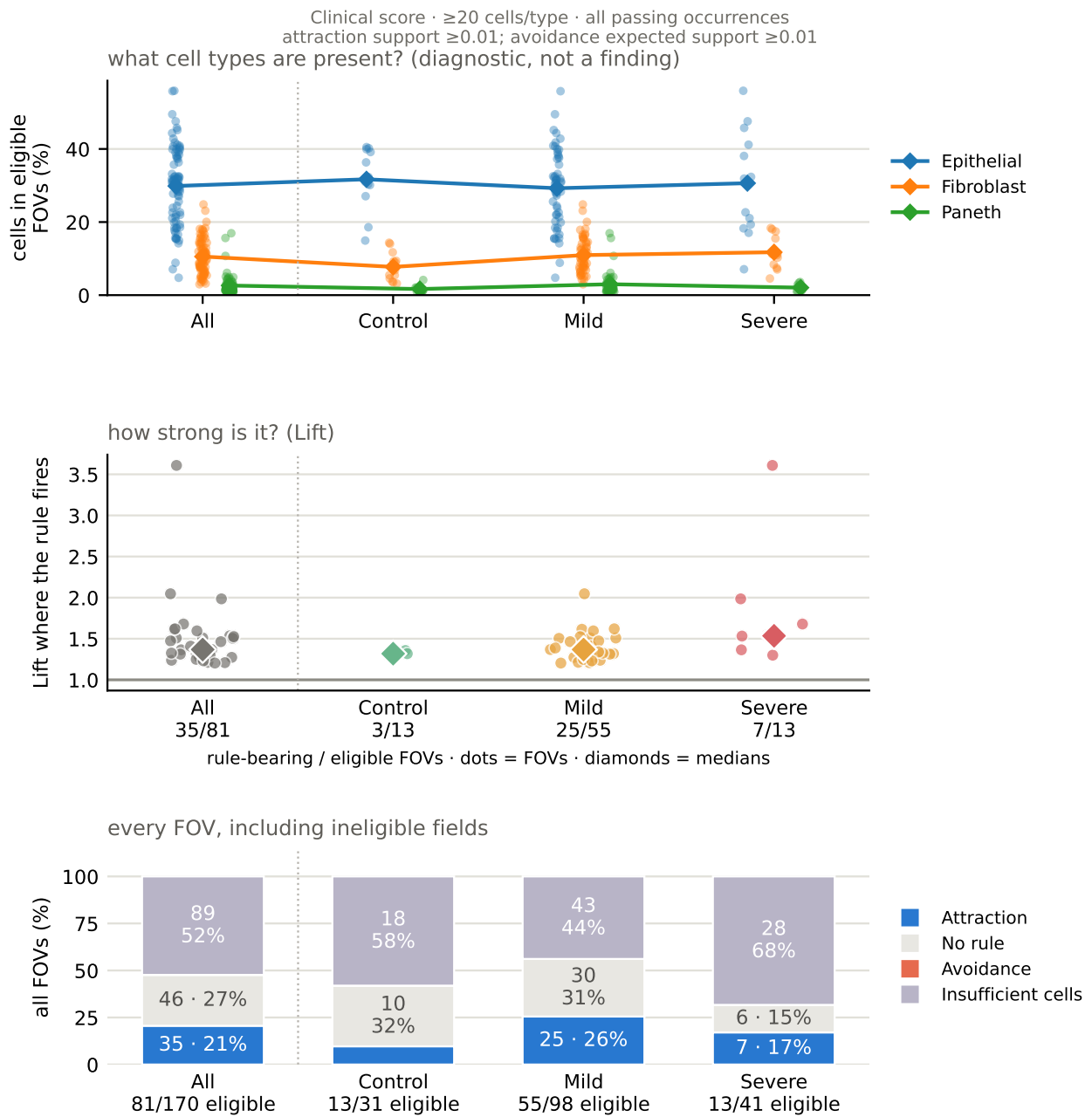


Figure 10: Fibroblast+Paneth → Epithelial occurs in 35/81 eligible FOVs under the all-passing endpoint.

Duodenum · Epithelial + Fibroblast → Paneth

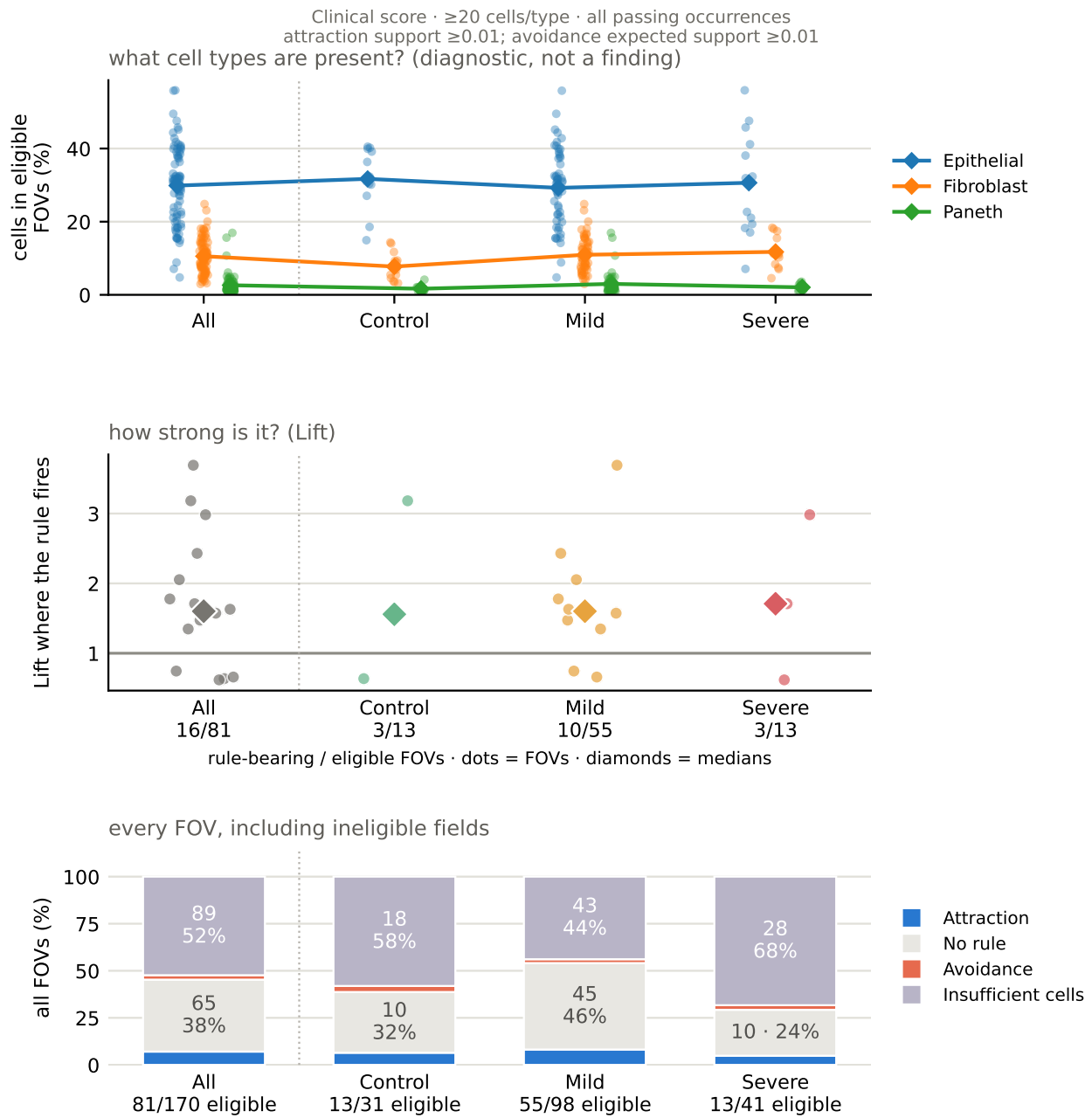


Figure 11: Epithelial+Fibroblast → Paneth occurs in 12/81. The patient-paired all-passing forward-minus-reverse difference is +26.4 percentage points ($q = 0.0038$); for informative-only rules it is -6.9 points ($q = 0.265$). This checks crypt organization, not molecular signaling direction.

2 Paper alignment

These rules were fixed from the [Azulay et al. \(2025\) GI aGVHD atlas](#) before inspecting the mined results. They are separate from automatic discovery. The atlas describes CD4 T cells and macrophages in distinct villus regions, and plasma cells in a band above the crypts; the rules test these broad spatial relationships, not exact cell subtypes.

Colon goblet avoidance of CD4T and macrophages weakens in mild disease and partly returns in severe disease.

Colon · Goblet → CD4T + Macrophage

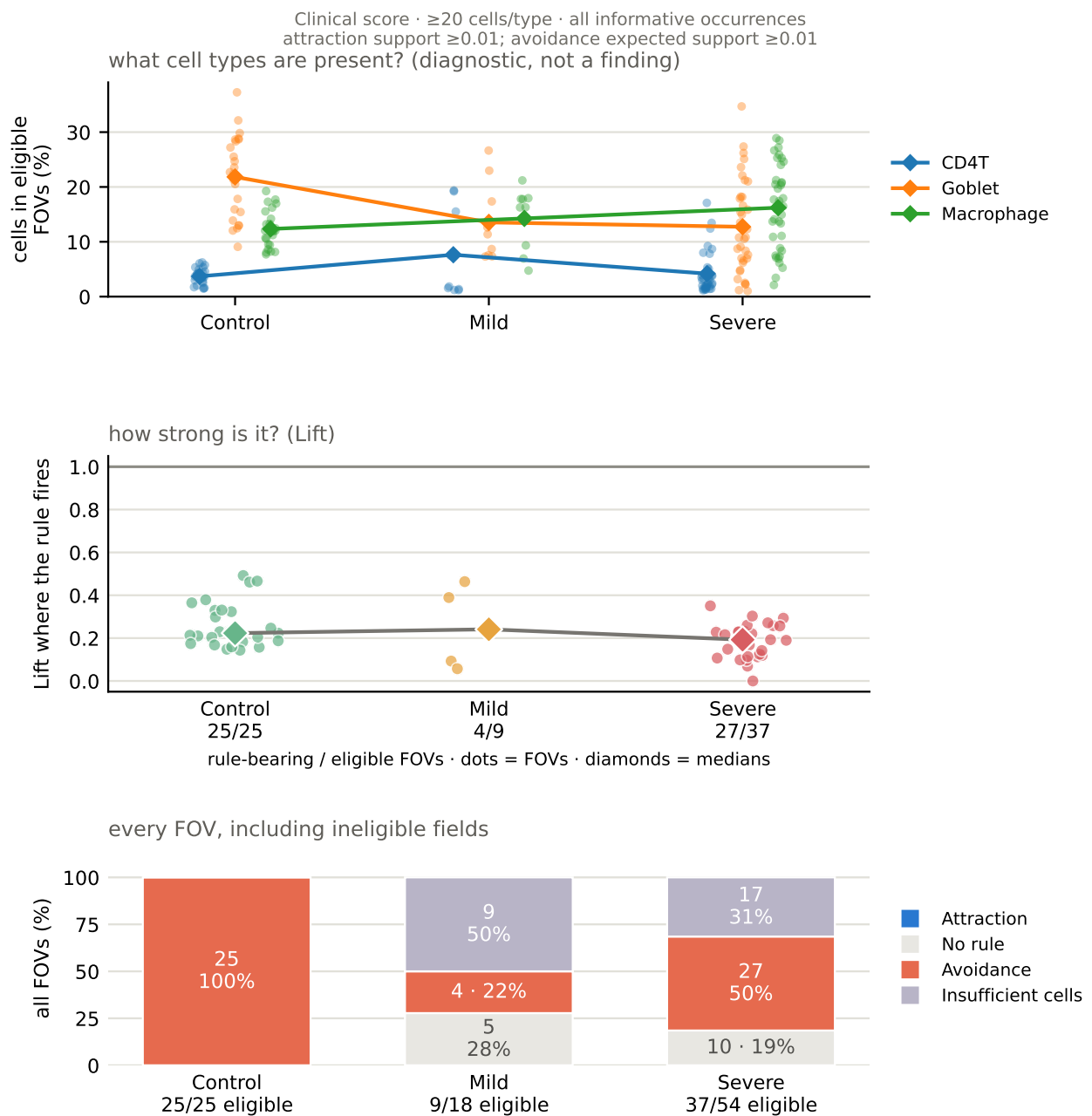


Figure 12: Goblet → CD4T+Macrophage: 25/25 control, 4/9 mild, 27/37 severe eligible FOVs; patient-stage $q = 0.674$. Here and below, only informative occurrences enter the rule-bearing counts; summary panels follow Figure 7.

Reversing the colon arrow gives the same curved stage shape at lower prevalence.

Colon · CD4T + Macrophage → Goblet

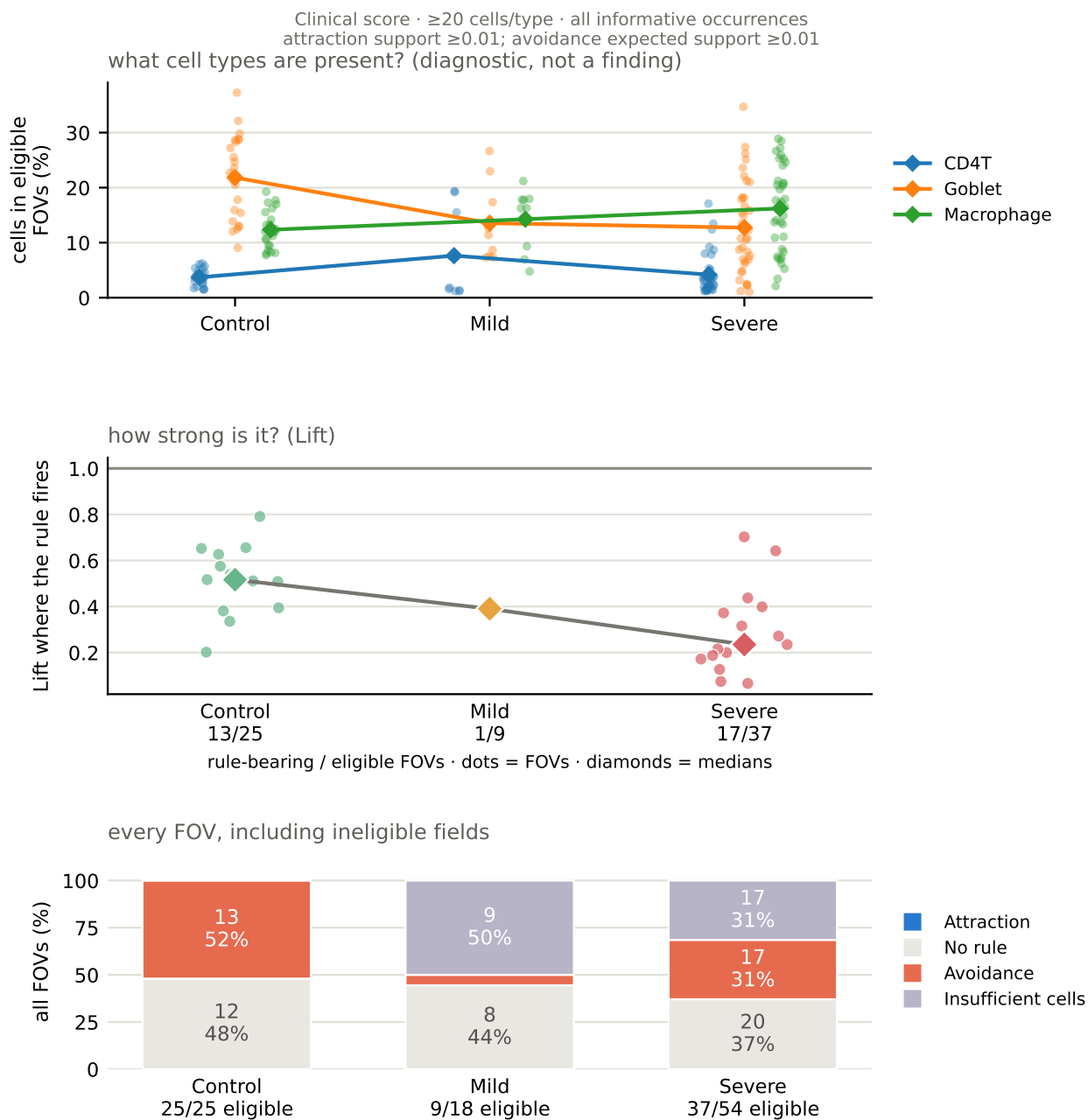


Figure 13: CD4T+Macrophage → Goblet: 13/25 control, 1/9 mild, 17/37 severe eligible FOVs. The two directions are measured separately.

The goblet-centred avoidance is visible in the colon tissue.

Colon · Goblet → CD4T + Macrophage

Avoidance examples · Control

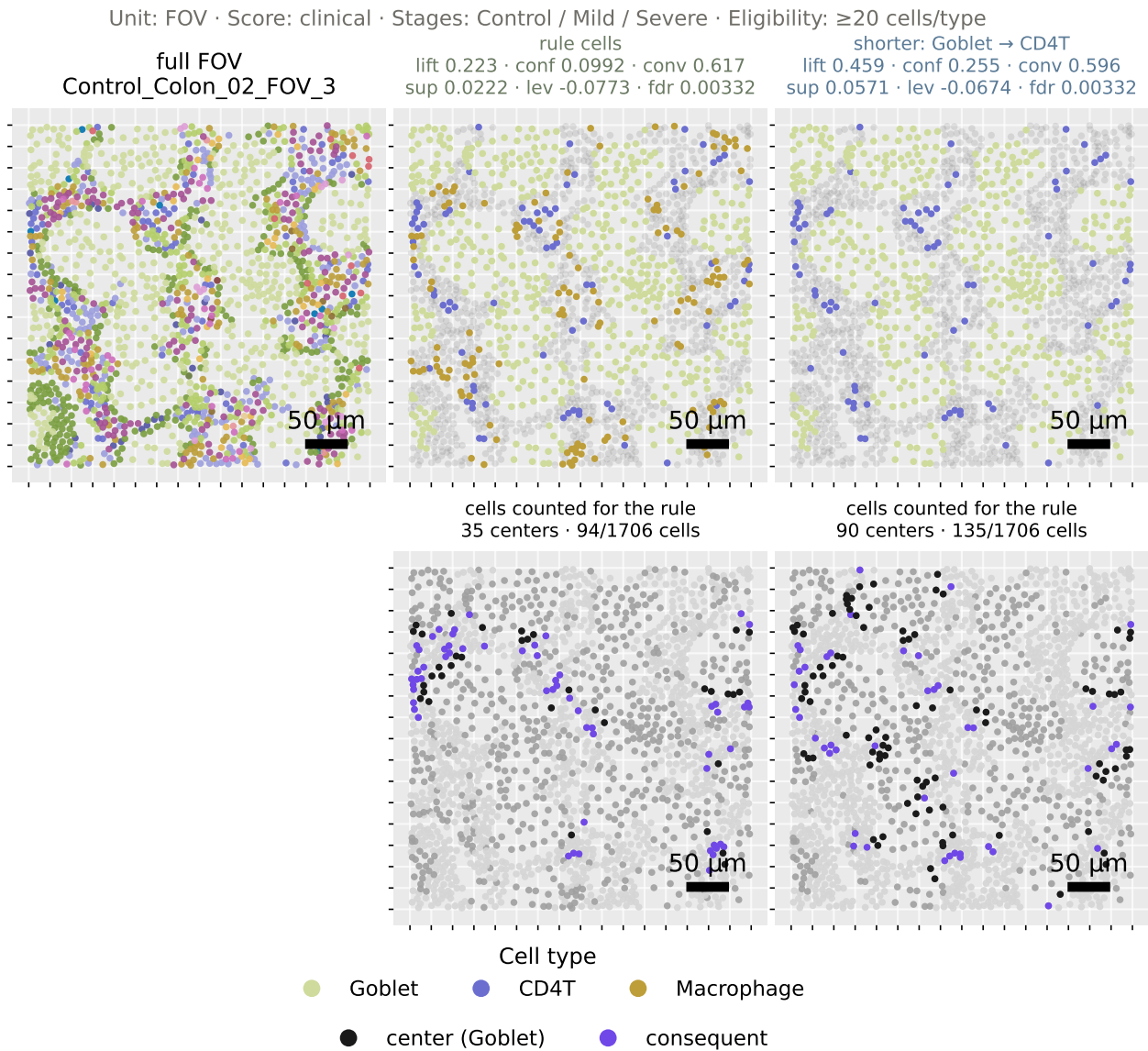


Figure 14: Representative map as in Figure 4.

In duodenum, goblet-centred avoidance becomes less common with severe disease.

Duodenum · Goblet → CD4T + Macrophage

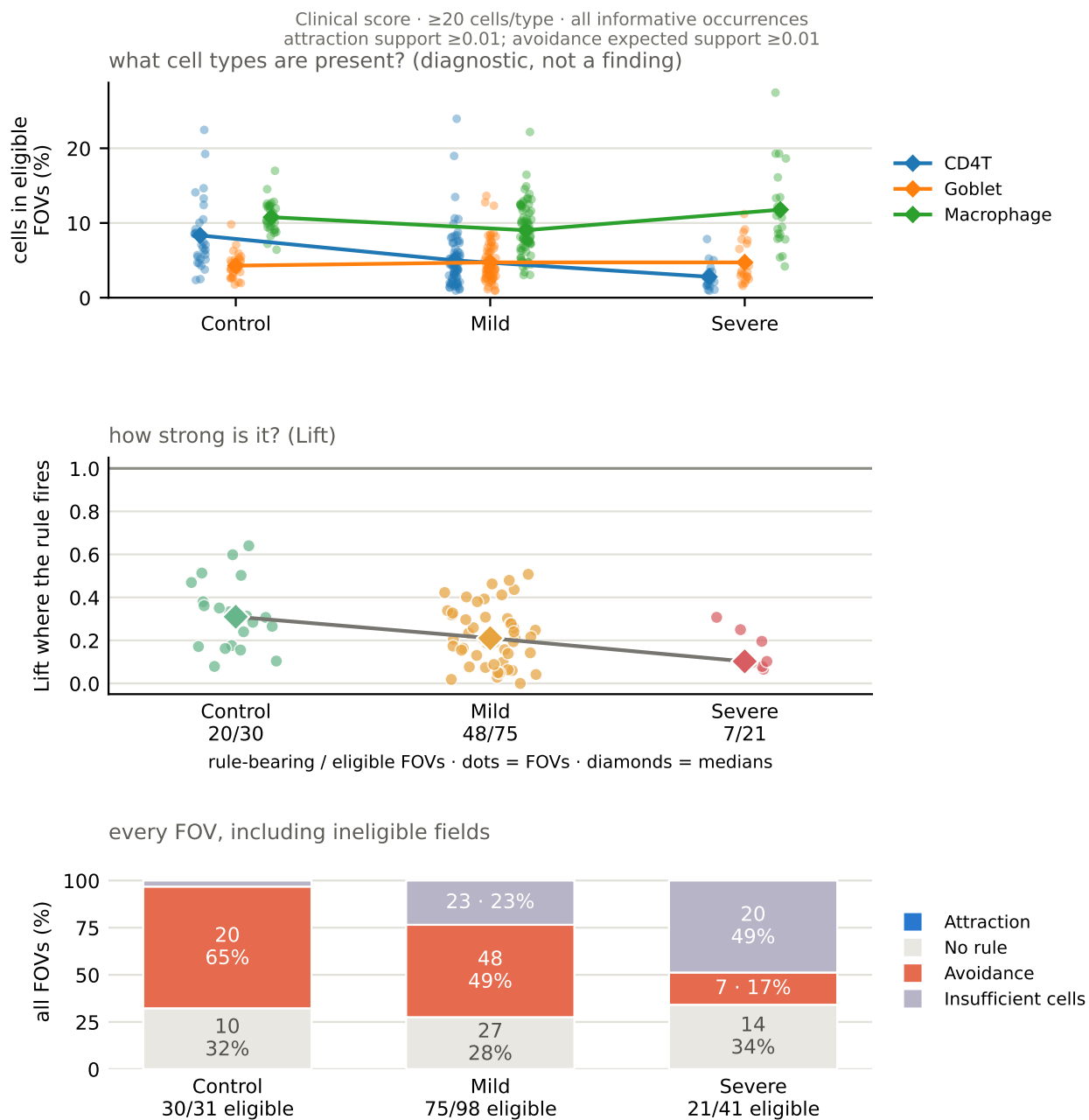


Figure 15: Goblet → CD4T+Macrophage: 20/30 control, 48/75 mild, 7/21 severe eligible FOVs; patient-stage $q = 0.704$.

The reverse duodenal arrow moves in the opposite visible direction.

Duodenum · CD4T + Macrophage → Goblet

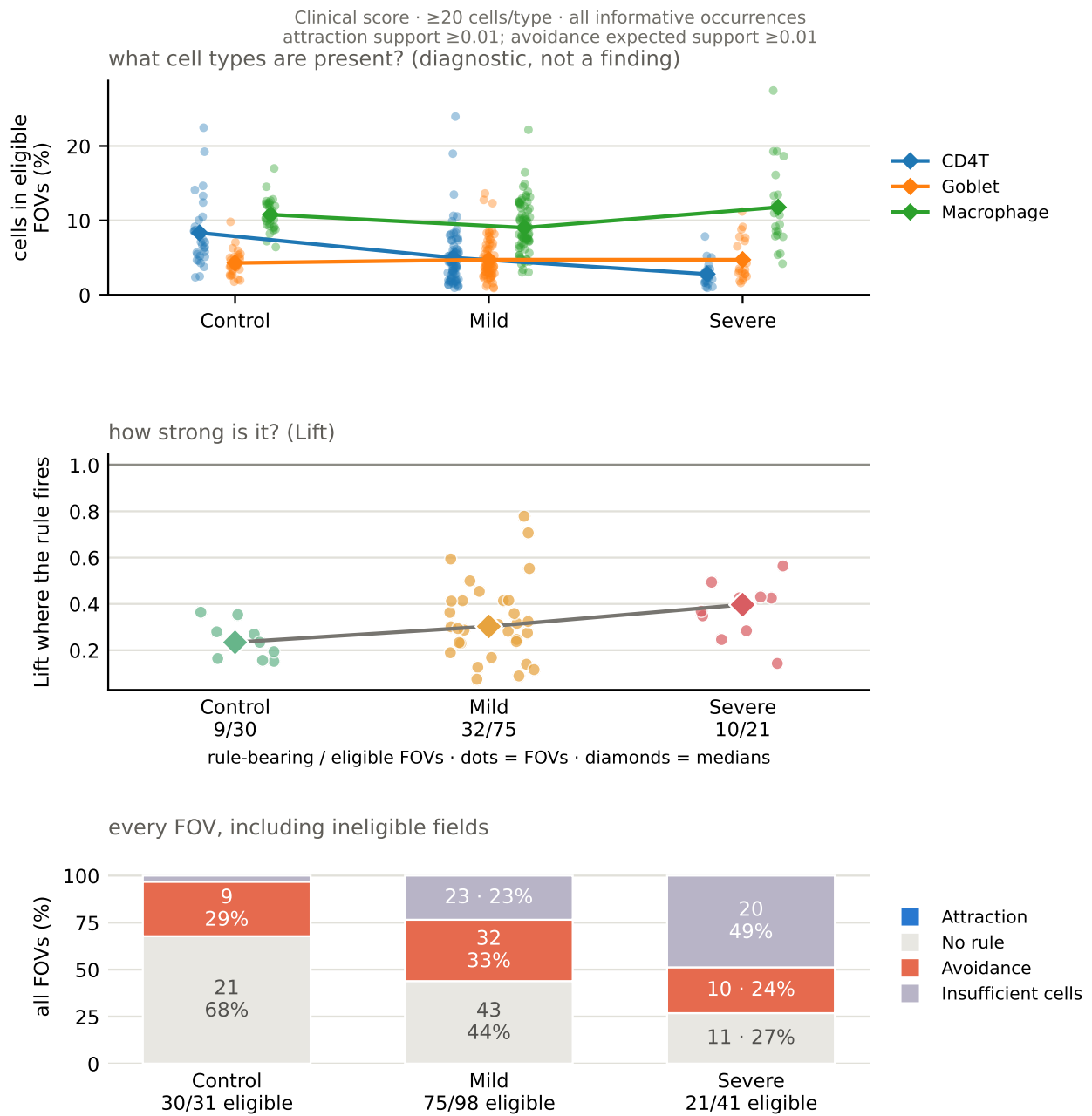


Figure 16: CD4T+Macrophage → Goblet: 9/30 control, 32/75 mild, 10/21 severe eligible FOVs; patient-stage $q = 0.927$. Direction is conditional association, not causality.

A plasma-centred duodenal attraction is most common in mild disease.

Duodenum · Plasma → CD4T + Epithelial

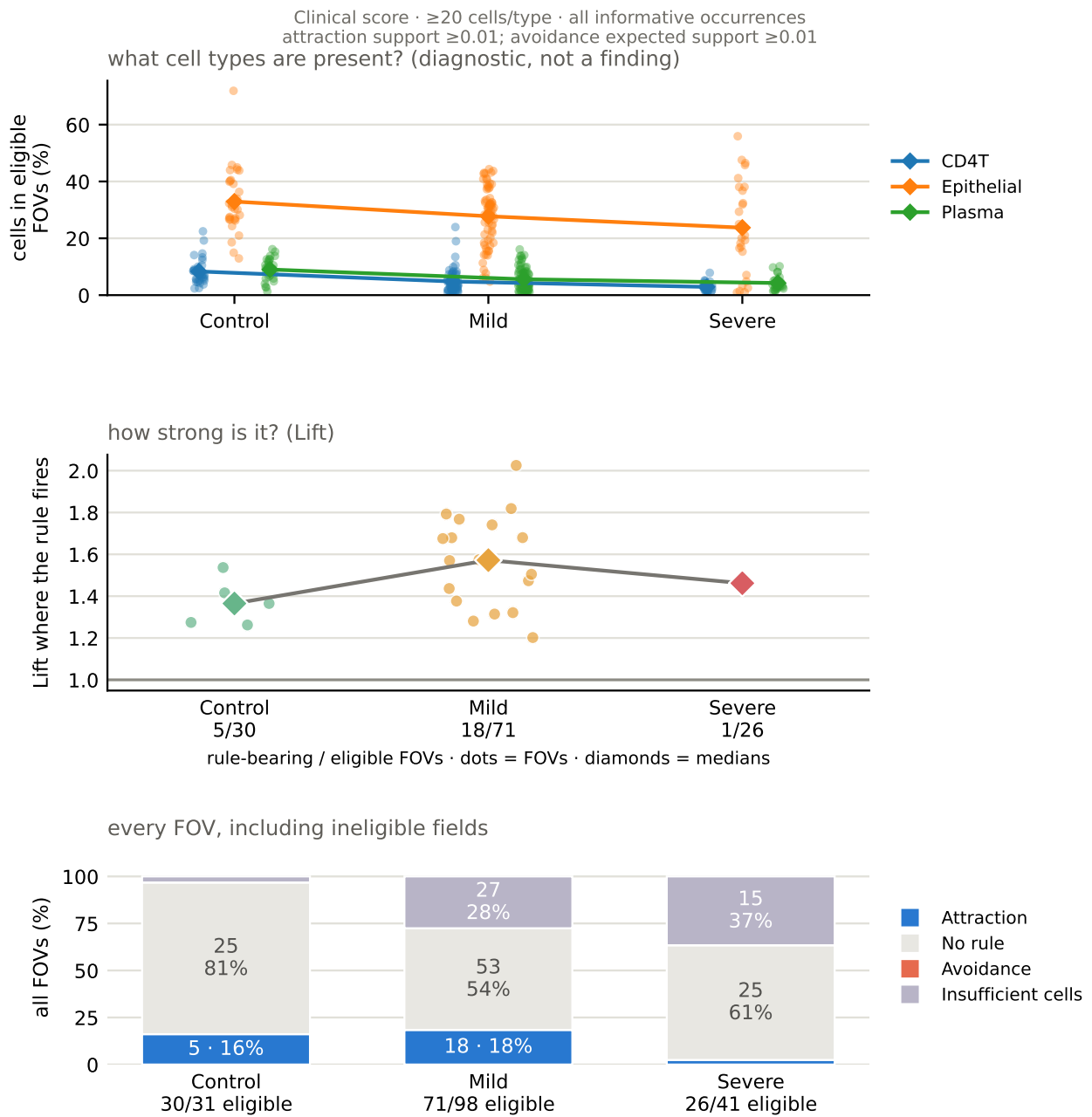


Figure 17: Plasma → CD4T+Epithelial: 5/30 control, 18/71 mild, 1/26 severe eligible FOVs; patient-stage $q = 0.785$. The atlas motivates the rule, but this stage contrast remains exploratory.

The plasma relationship can be inspected in its tissue setting.

Duodenum · Plasma → CD4T + Epithelial

Attraction examples · Mild

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

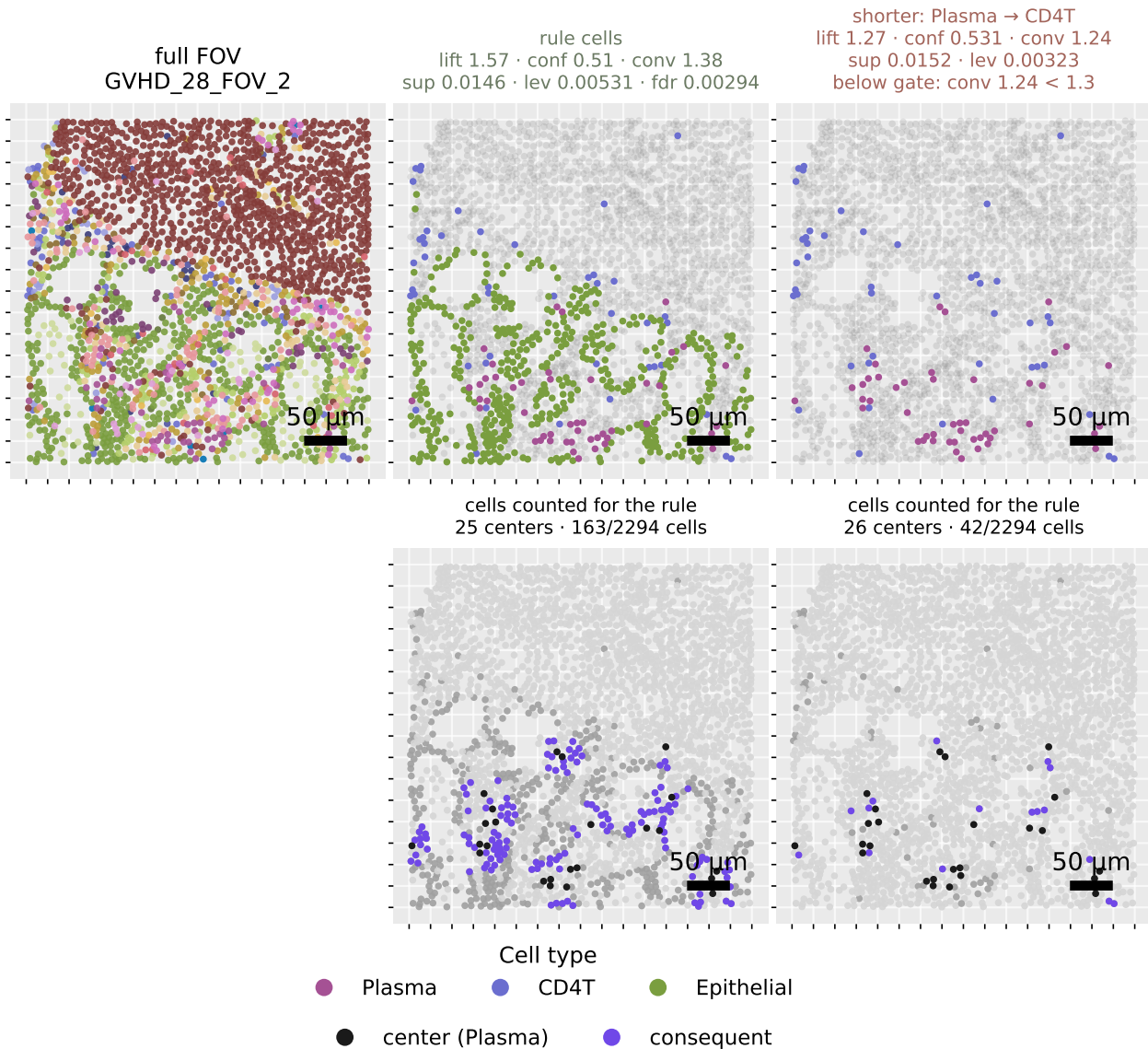


Figure 18: Representative map as in Figure 4.

3 New findings

All rules here first appeared in automatic, configurable visible-effect screens. The screen required adequate eligible fields and recurring rule-bearing patients, then ranked prevalence, conditional Lift, or measured parent–complex trend gaps. Literature is used only to interpret selected rules. The figures show exploratory effects with their denominators; no new result below has a patient-level $q < 0.05$.

3.1 Colon goblet rules separate cell contexts

A four-type goblet avoidance nearly disappears from control to severe colon.

Colon · Goblet → CD8T + Fibroblast + Plasma

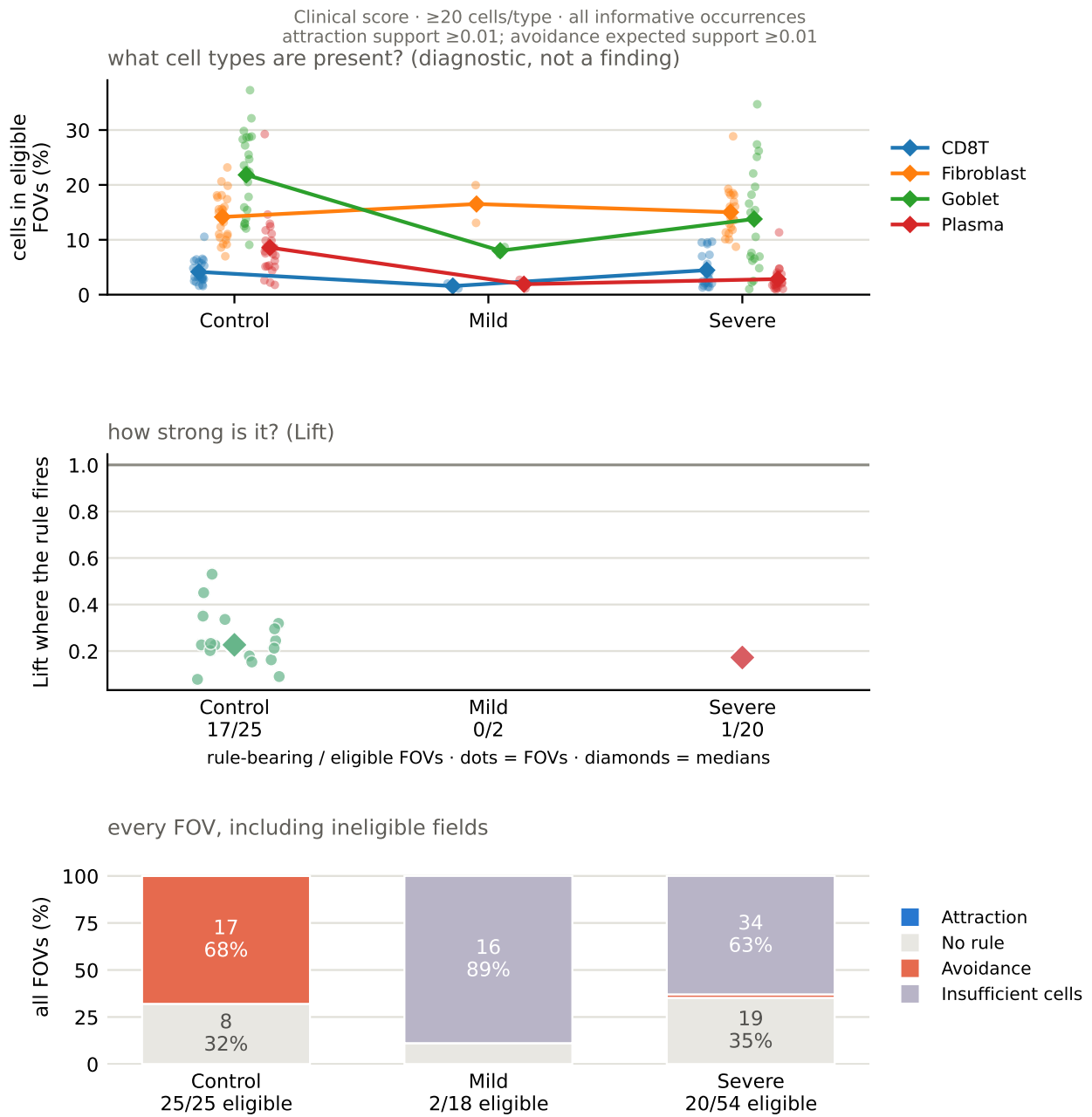


Figure 19: Goblet → CD8T+Fibroblast+Plasma: 17/25 control and 1/20 severe eligible FOVs, in five rule-bearing patients overall. This was the top four-type visible-gap result; patient-stage $q = 0.350$. Only 2 mild FOVs are eligible.

Its shorter goblet-to-CD8T-and-plasma rule stays common when the four-type rule falls.

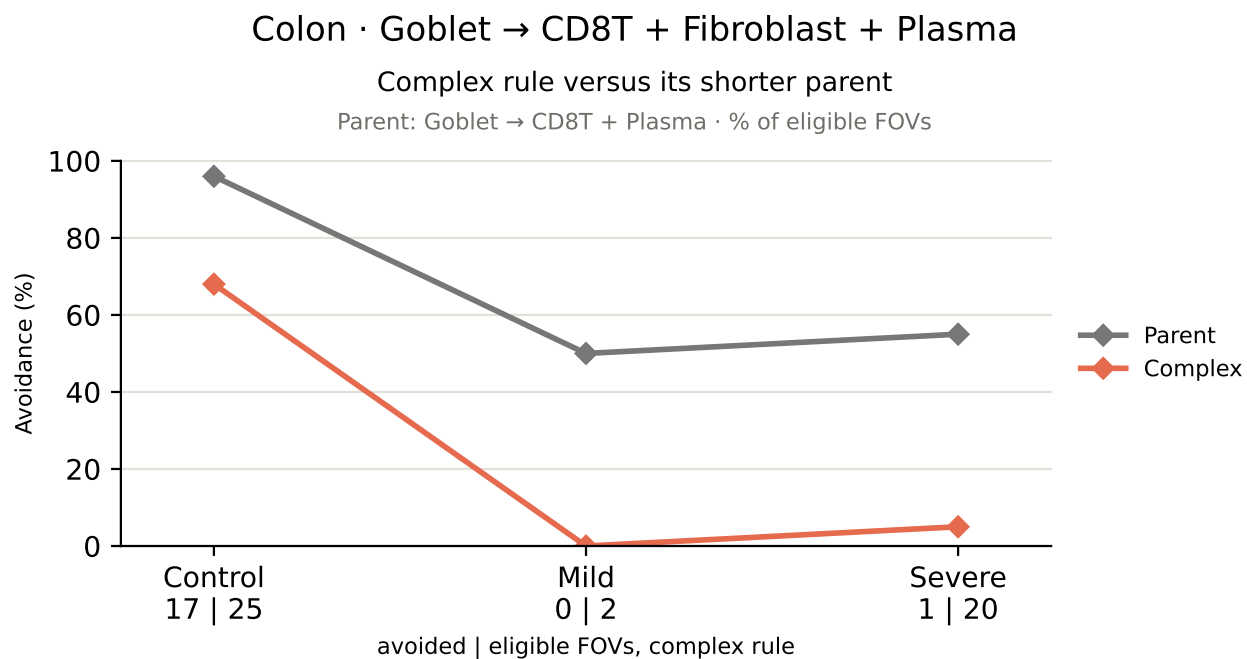


Figure 20: The measured shorter rule is 24/25 control and 11/20 severe eligible FOVs, compared with 17/25 and 1/20 for the complex avoidance. Same eligibility and prevalence calculation as Figure 8.

The control fields show the four cell types behind the goblet avoidance.

Colon · Goblet → CD8T + Fibroblast + Plasma

Avoidance examples · Control

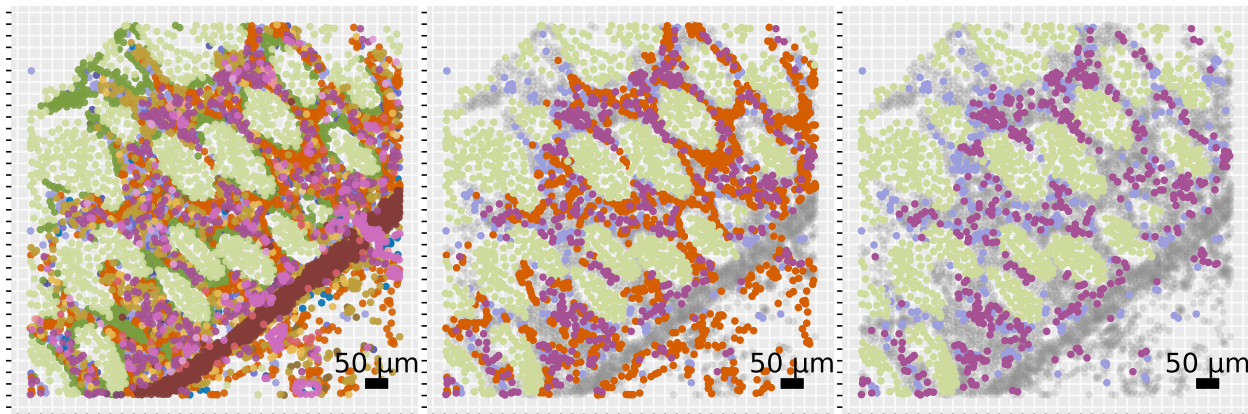
Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

shorter: Goblet → CD8T + Plasma

full FOV
Control_Colon_02_FOV_6

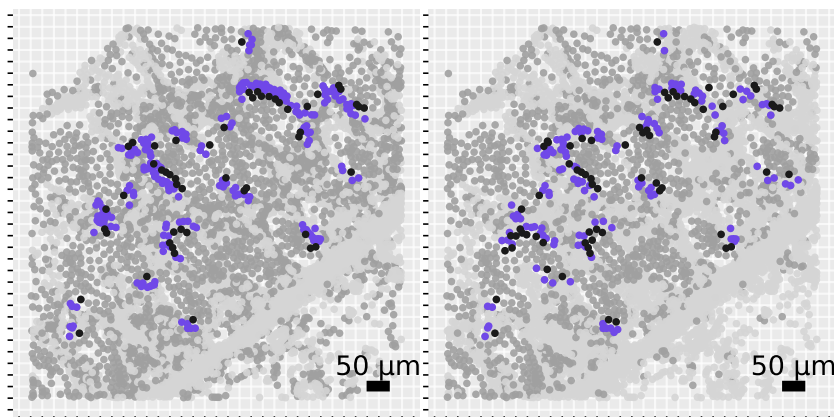
rule cells
lift 0.227 · conf 0.092 · conv 0.654
sup 0.01 · lev -0.0341 · fdr 0.00206

lift 0.286 · conf 0.124 · conv 0.648
sup 0.0135 · lev -0.0336 · fdr 0.00206



cells counted for the rule
52 centers · 258/5709 cells

cells counted for the rule
70 centers · 209/5709 cells



Cell type

- Goblet
- CD8T
- Fibroblast
- Plasma
- center (Goblet)
- consequent

Figure 21: Representative control maps as in Figure 4.

Eligible severe fields without the four-type avoidance still contain its cell types.

Colon · Goblet → CD8T + Fibroblast + Plasma

Fields without the rule

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

```
rule cells
lift 0.294 · conf 0.105 · conv 0.718
sup 0.0155 · lev -0.0372 · fdr 0.00207
measured here, dropped by a later
filter
```

cells counted for the rule
78 centers · 354/5691 cells

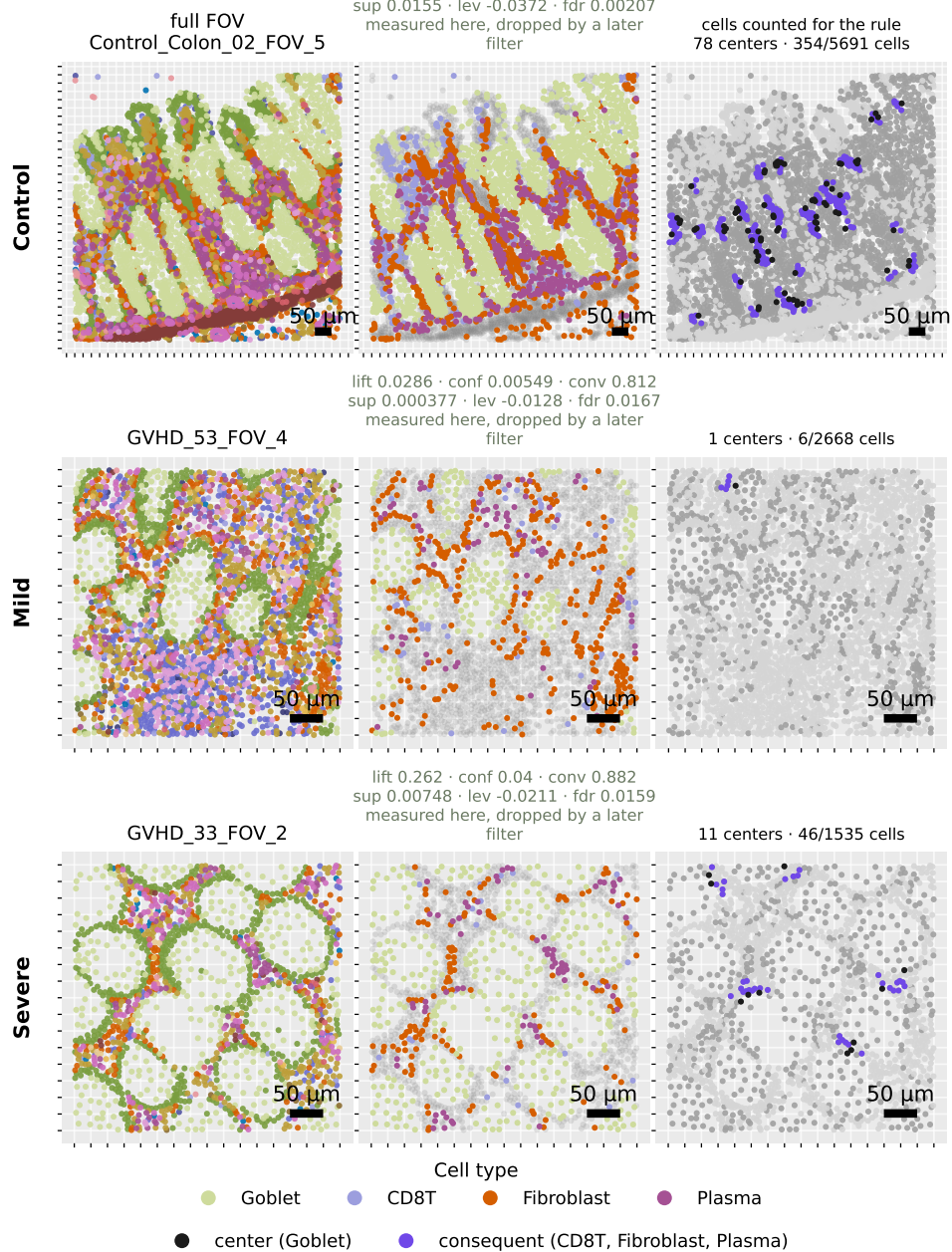


Figure 22: No-rule FOVs pass the same cell-count eligibility but have no stored occurrence passing this investigation filter; cell presence alone does not imply a rule.

A vascular goblet avoidance shows a second large control-to-severe contrast.

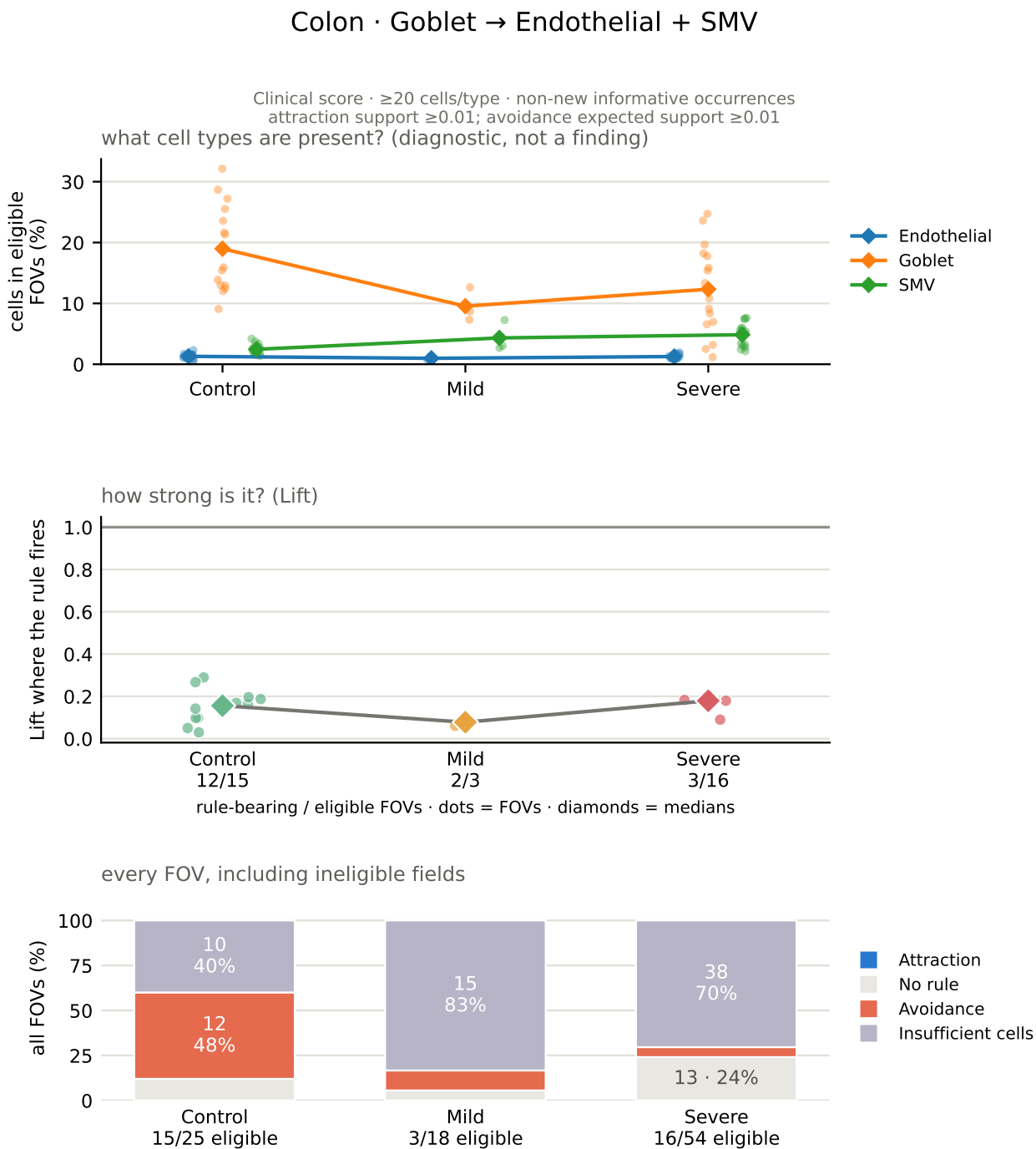


Figure 23: Goblet → Endothelial+SMV: 12/15 control versus 3/16 severe eligible FOVs, with nine rule-bearing patients overall; patient-stage $q = 0.478$ in the three-type screen. Mild has only 3 eligible FOVs.

The vascular avoidance is visible across whole control fields

Colon · Goblet → Endothelial + SMV

Avoidance examples · Control

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

rule cells

shorter: Goblet → Endothelial

full FOV

lift 0.164 · conf 0.0161 · conv 0.916
sup 0.00297 · lev -0.0152 · fdr 0.00374

lift 0.36 · conf 0.0764 · conv 0.853
sup 0.014 · lev -0.025 · fdr 0.00196

Control_Colon_01_FOV_6

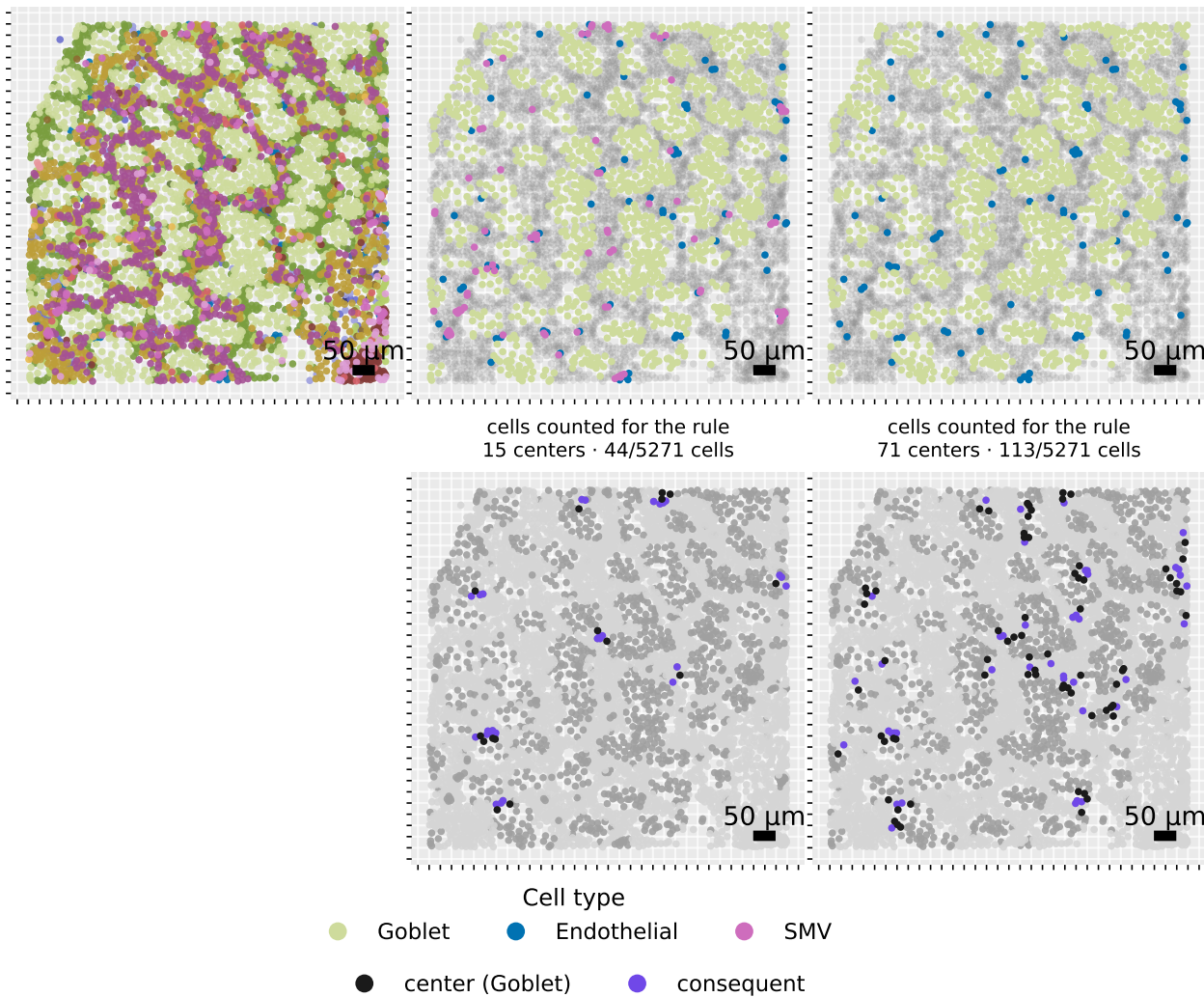


Figure 24: Representative maps as in Figure 4.

A different goblet avoidance remains more common in severe colon.

Colon · Goblet → CD8T + Fibroblast

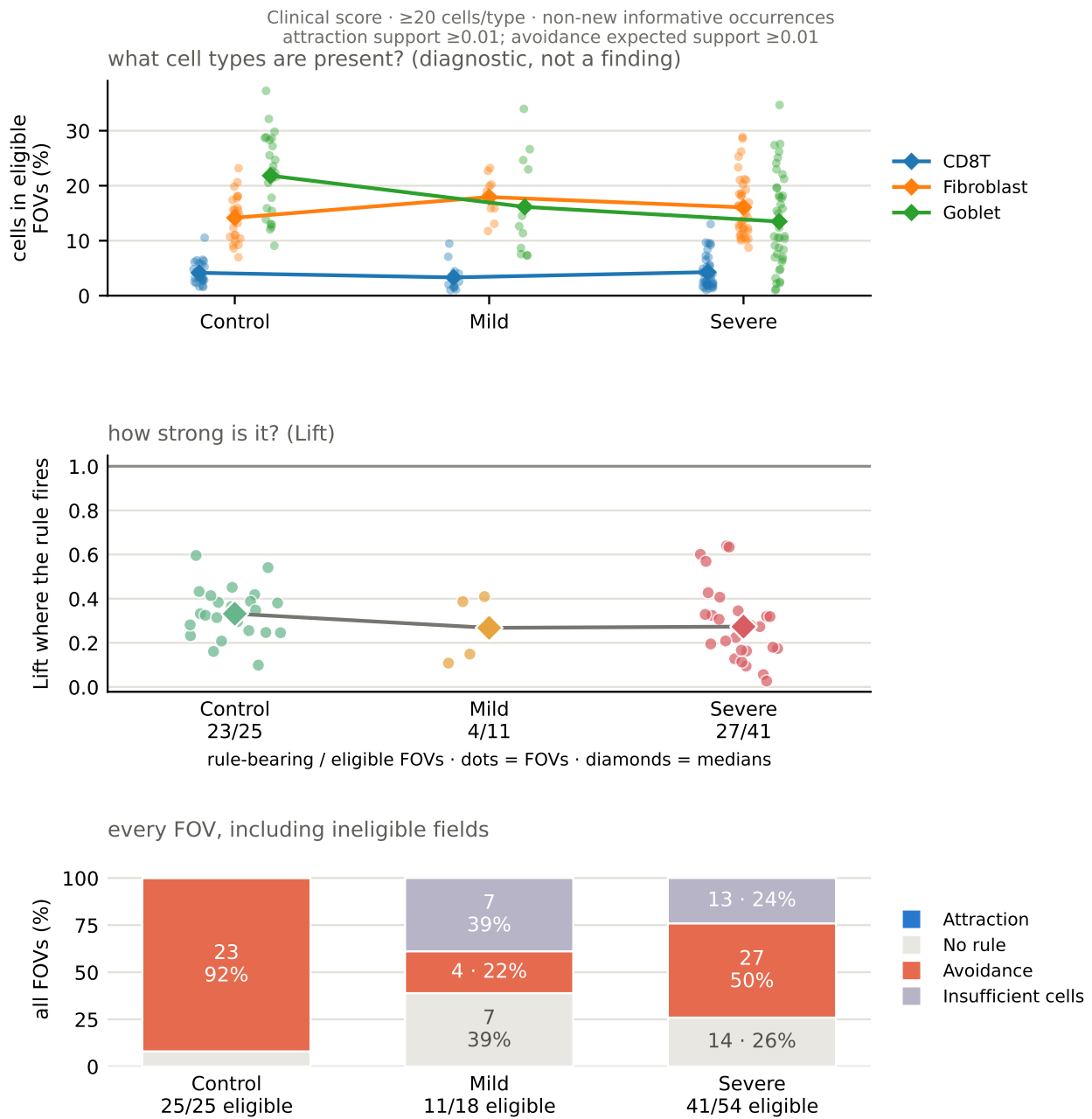


Figure 25: Goblet → CD8T+Fibroblast: 23/25 control and 27/41 severe eligible FOVs, 17 rule-bearing patients overall; patient-stage $q = 0.488$. Its distinct denominator shows why this three-type rule should not be conflated with Figure 19.

3.2 Colon plasma attraction loses a cell context

The atlas reports disrupted plasma-cell organization in aGVHD (Azulay et al., 2025); it does not specify this conditional spatial rule.

A new-class epithelial-fibroblast attraction to plasma is common in controls and rare in severe colon.

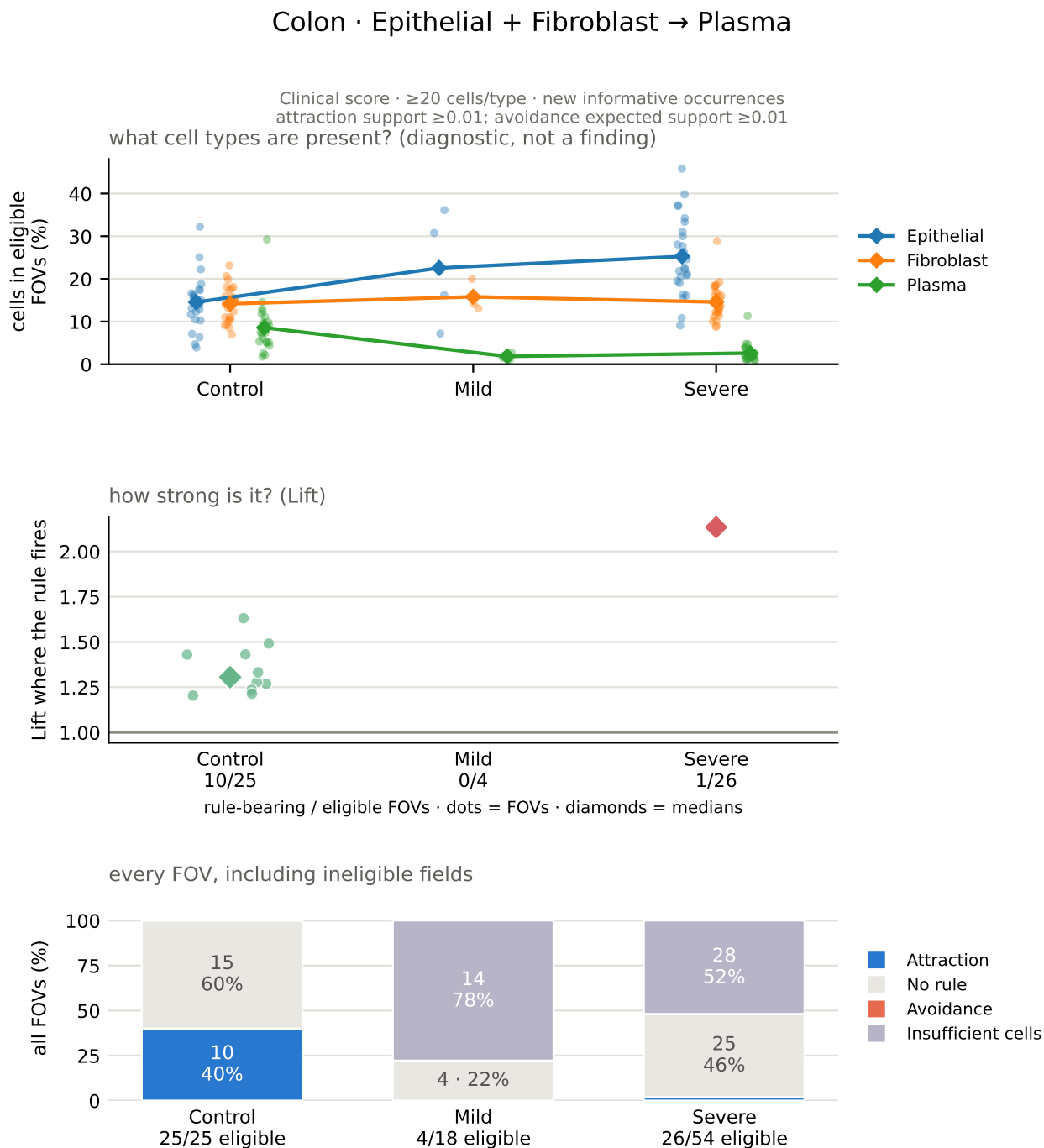


Figure 26: Epithelial+Fibroblast → Plasma: 10/25 control versus 1/26 severe eligible FOVs in the automatically selected *new-class* stream, five rule-bearing patients overall; patient-stage $q = 0.658$. New-class means no shorter stored rule explains the occurrence.

The plasma-cell locations can be seen in representative full fields.

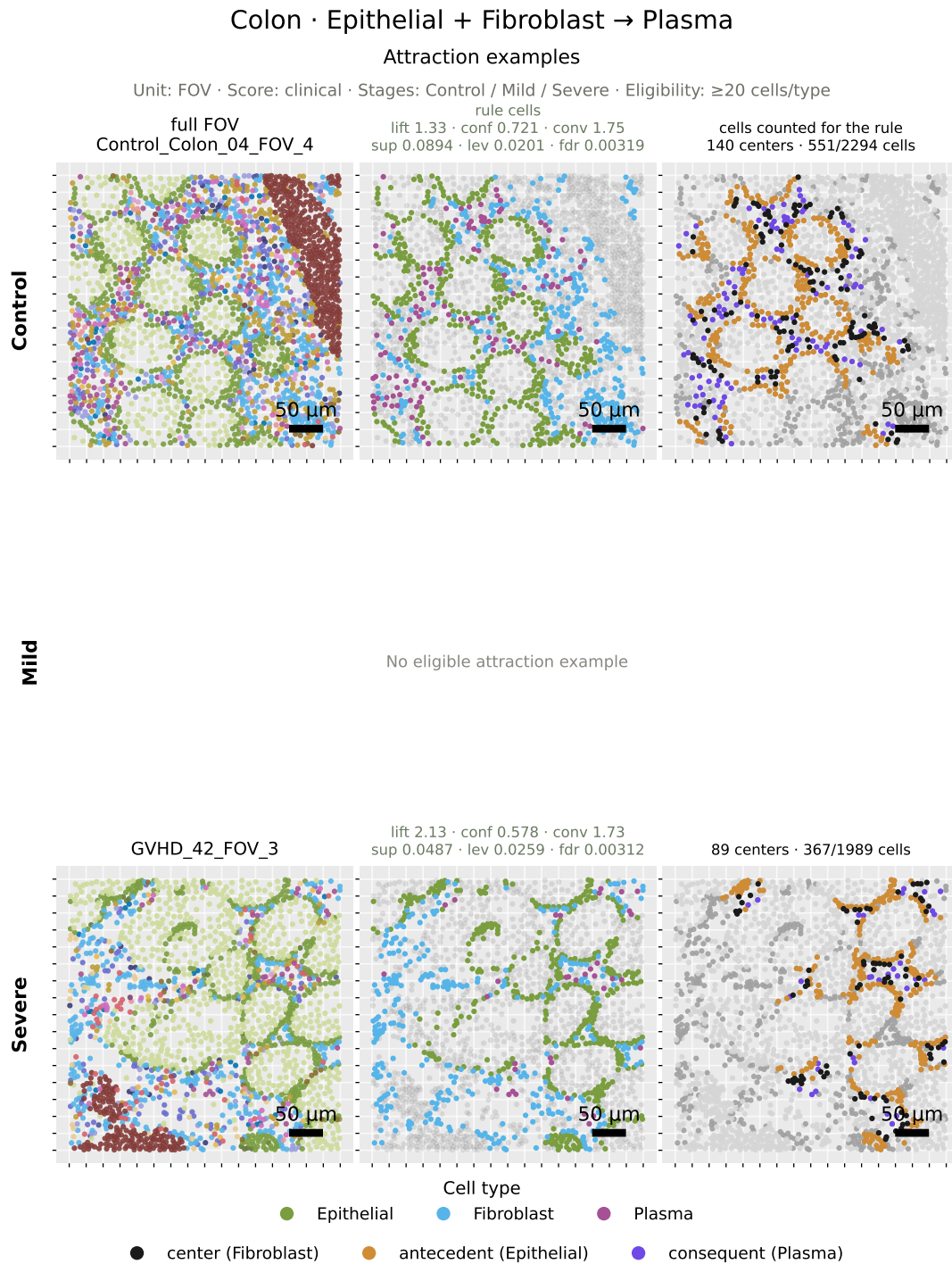


Figure 27: Representative maps as in Figure 4.

3.3 Duodenal goblet avoidance gains fibroblast context

Sensory neurons can regulate goblet secretion in other settings (Yang et al., 2022); this broad spatial rule does not test that signaling mechanism.

Duodenum · Goblet → Fibroblast + Neuron

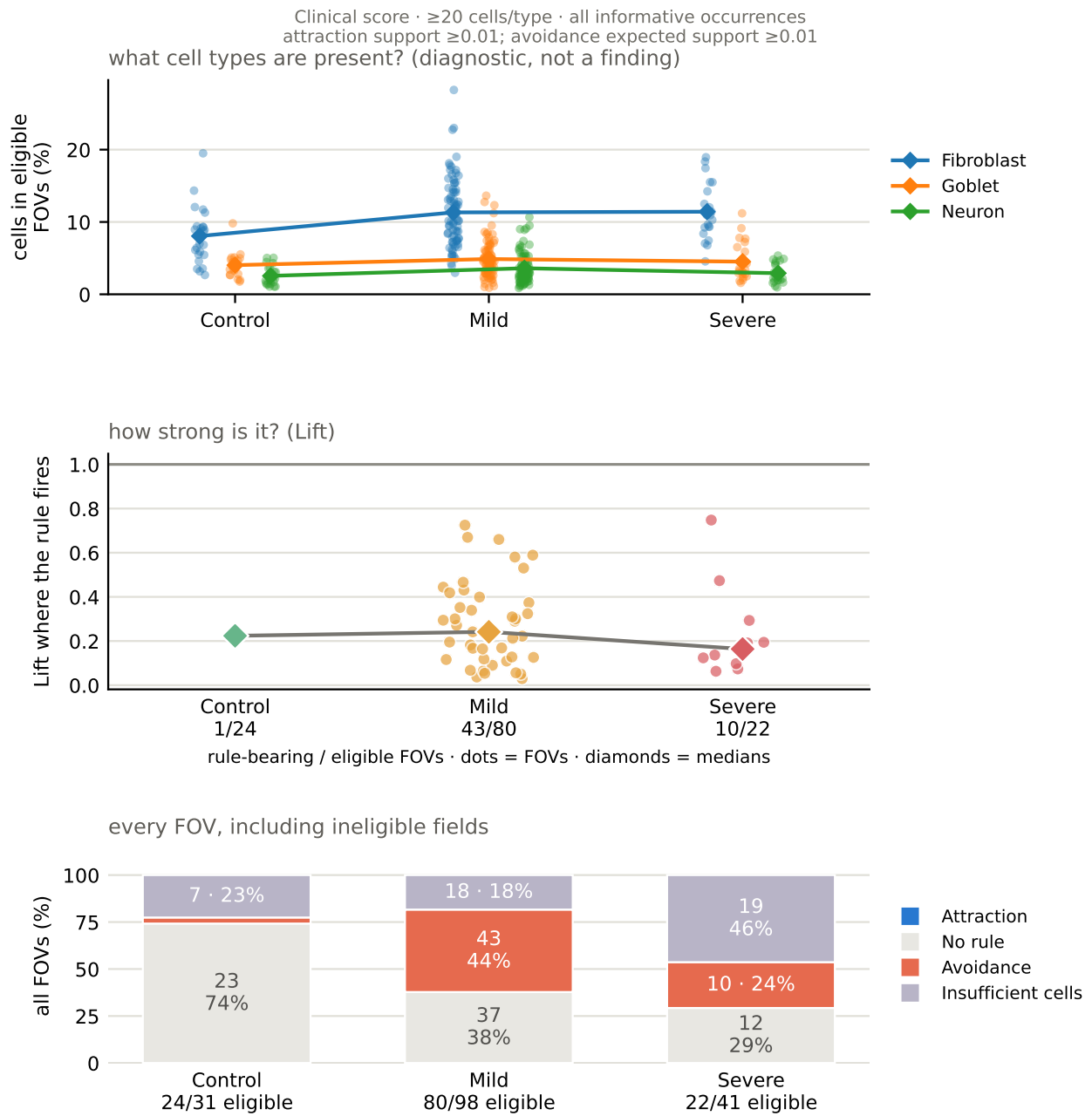


Figure 28: Goblet → Fibroblast+Neuron avoidance appears in 1/24 control, 43/80 mild, and 10/22 severe eligible FOVs, across 1/9, 21/26, and 6/7 eligible patients respectively. It entered the automatic up-to-four-type screen through a visible gap; the patient-level stage test remains exploratory ($q = 0.350$). Summary panels follow Figure 7.

The shorter goblet-to-neuron avoidance is more common throughout, so fibroblast context sharpens the stage contrast.

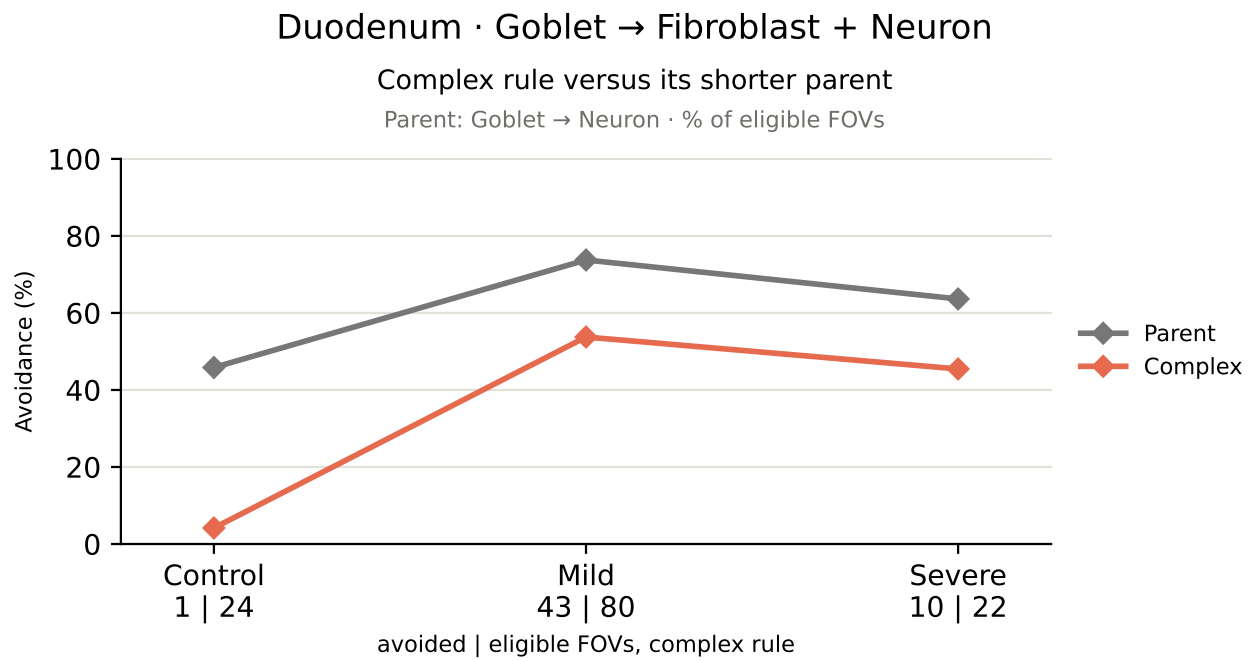


Figure 29: The measured Goblet → Neuron parent and the complex rule use the same eligible FOVs. Prevalence is calculated as in Figure 8; adding fibroblast lowers control prevalence much more than disease prevalence.

One mild-disease field shows the complex avoidance and its measured shorter rule in tissue.

Duodenum · Goblet → Fibroblast + Neuron

Avoidance examples · Mild

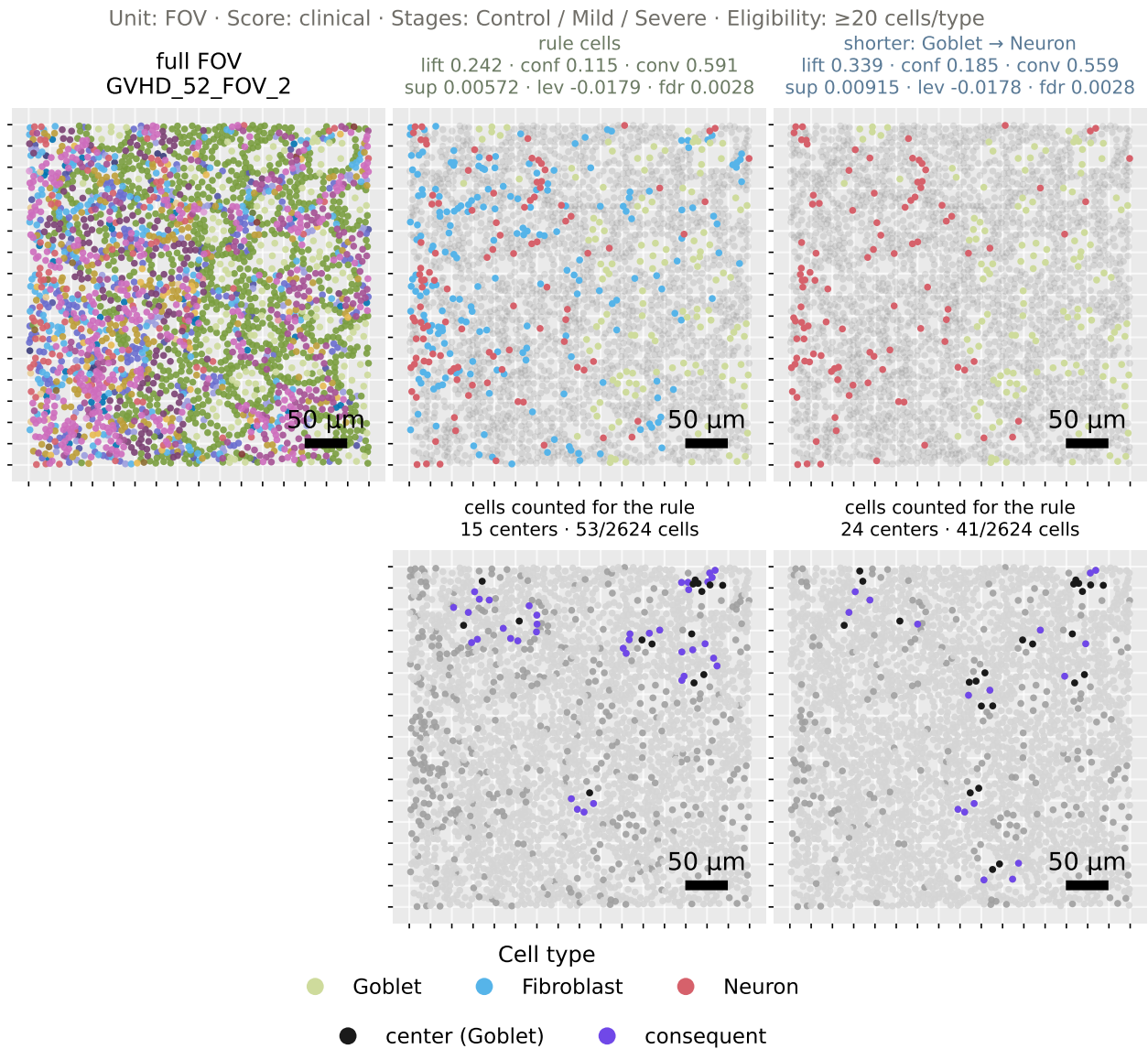


Figure 30: The top row shows the full FOV and the cell types named by each rule; the lower panels highlight centres and consequent cells counted for the complex and shorter rules. This automatically chosen field is near the mild-stage median Lift and does not by itself establish the stage difference.

3.4 Duodenal vessel and crypt patterns

Intestinal aGVHD can involve endothelial injury (Cordes et al., 2021); that work does not predict this muscle-conditioned arrow.

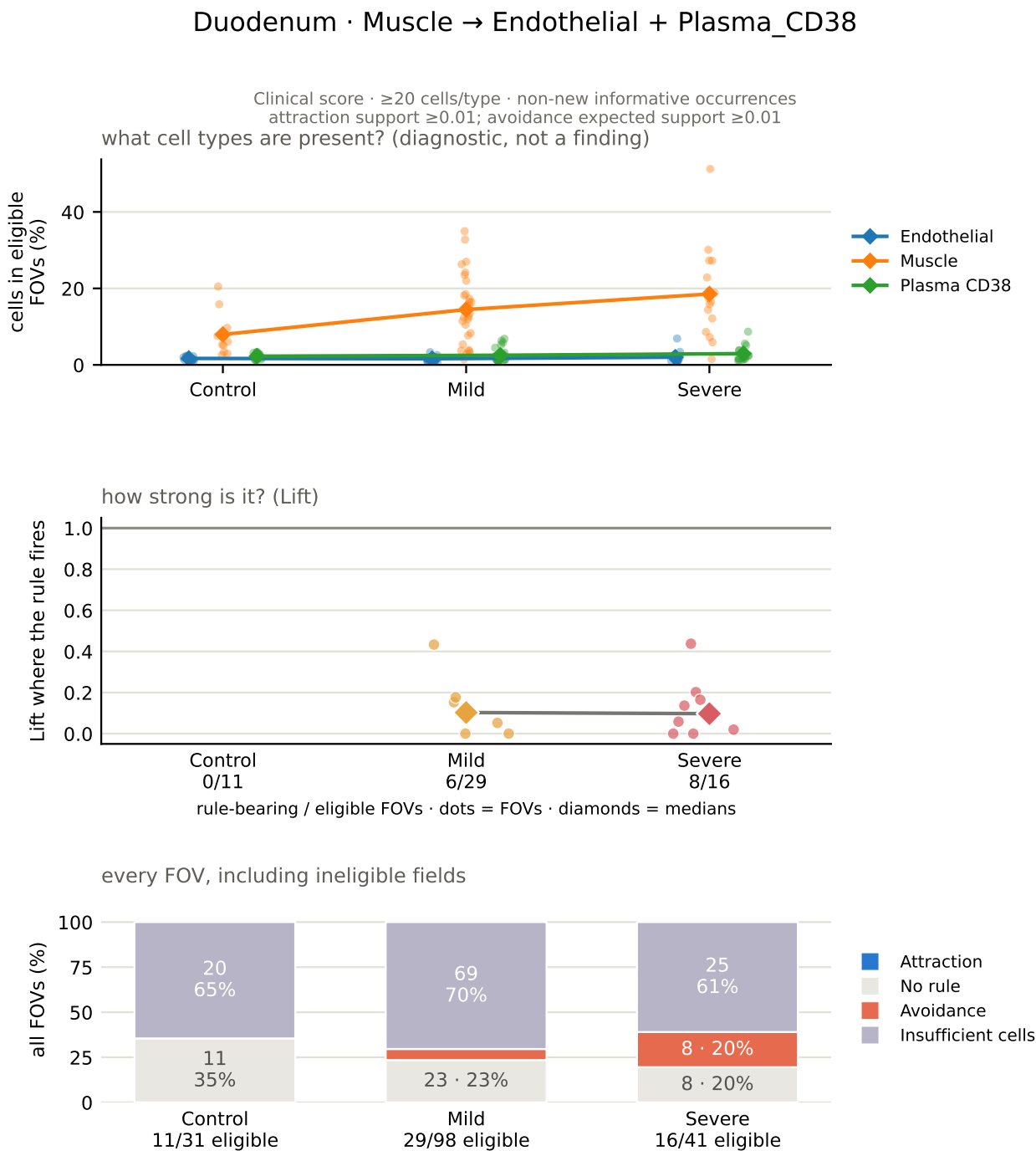


Figure 31: Muscle → Endothelial+Plasma_CD38: 0/11 control, 6/29 mild, and 8/16 severe eligible FOVs, ten rule-bearing patients overall; patient-stage $q = 0.600$ in the three-type screen.

Its shorter muscle-to-plasma rule is already present in controls.

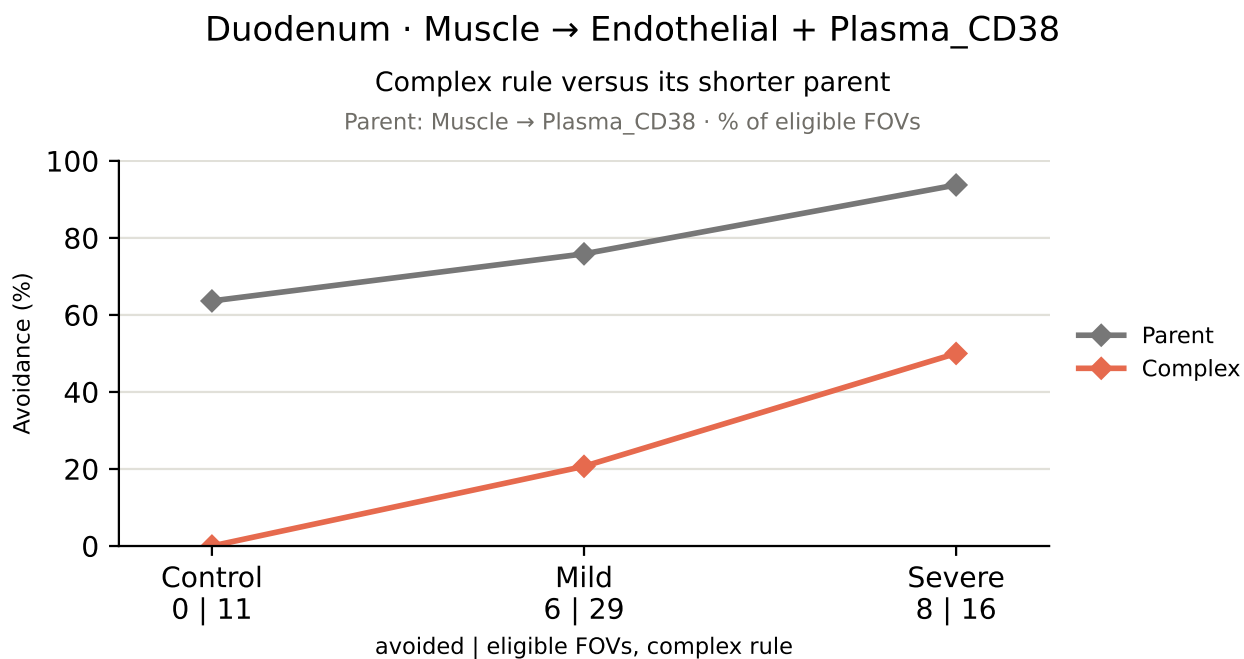


Figure 32: Muscle → Plasma_CD38 avoidance is 7/11 control and 15/16 severe eligible FOVs. The comparison uses the same FOV denominator as the complex rule.

The muscle, vessel and plasma labels are visible together in severe duodenal fields.

Duodenum · Muscle → Endothelial + Plasma_CD38

Avoidance examples · Severe

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

full FOV
GVHD_24_FOV_2

rule cells
lift 0.0583 · conf 0.0103 · conv 0.832
sup 0.00299 · lev -0.0484 · fdr 0.00218

shorter: Muscle → Plasma_CD38
lift 0.399 · conf 0.123 · conv 0.788
sup 0.0359 · lev -0.054 · fdr 0.00218

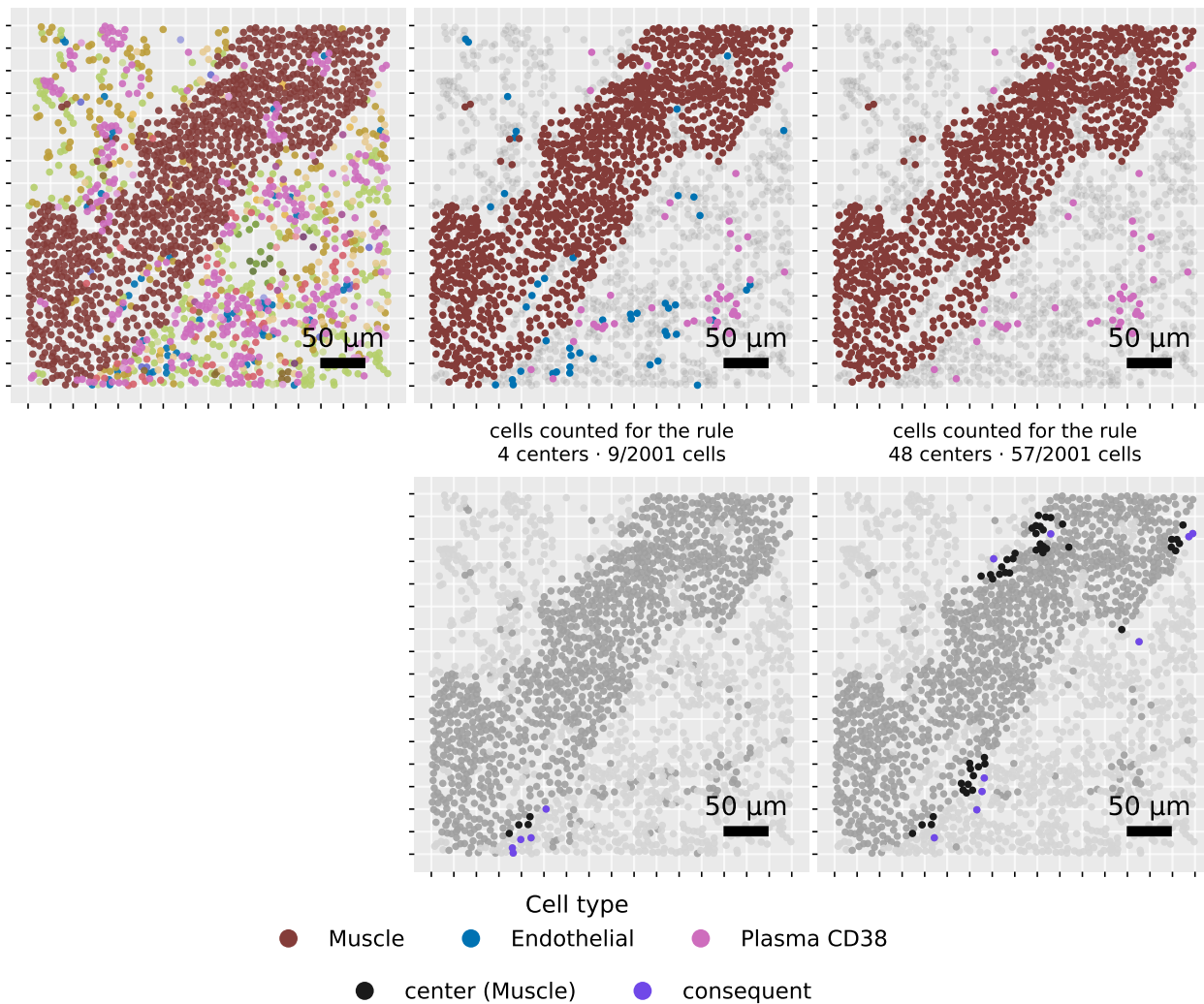


Figure 33: Representative maps as in Figure 4.

A four-type Paneth-related attraction appears frequently in severe duodenum at the baseline gate.

Duodenum · Fibroblast + Paneth → Epithelial + Neuron

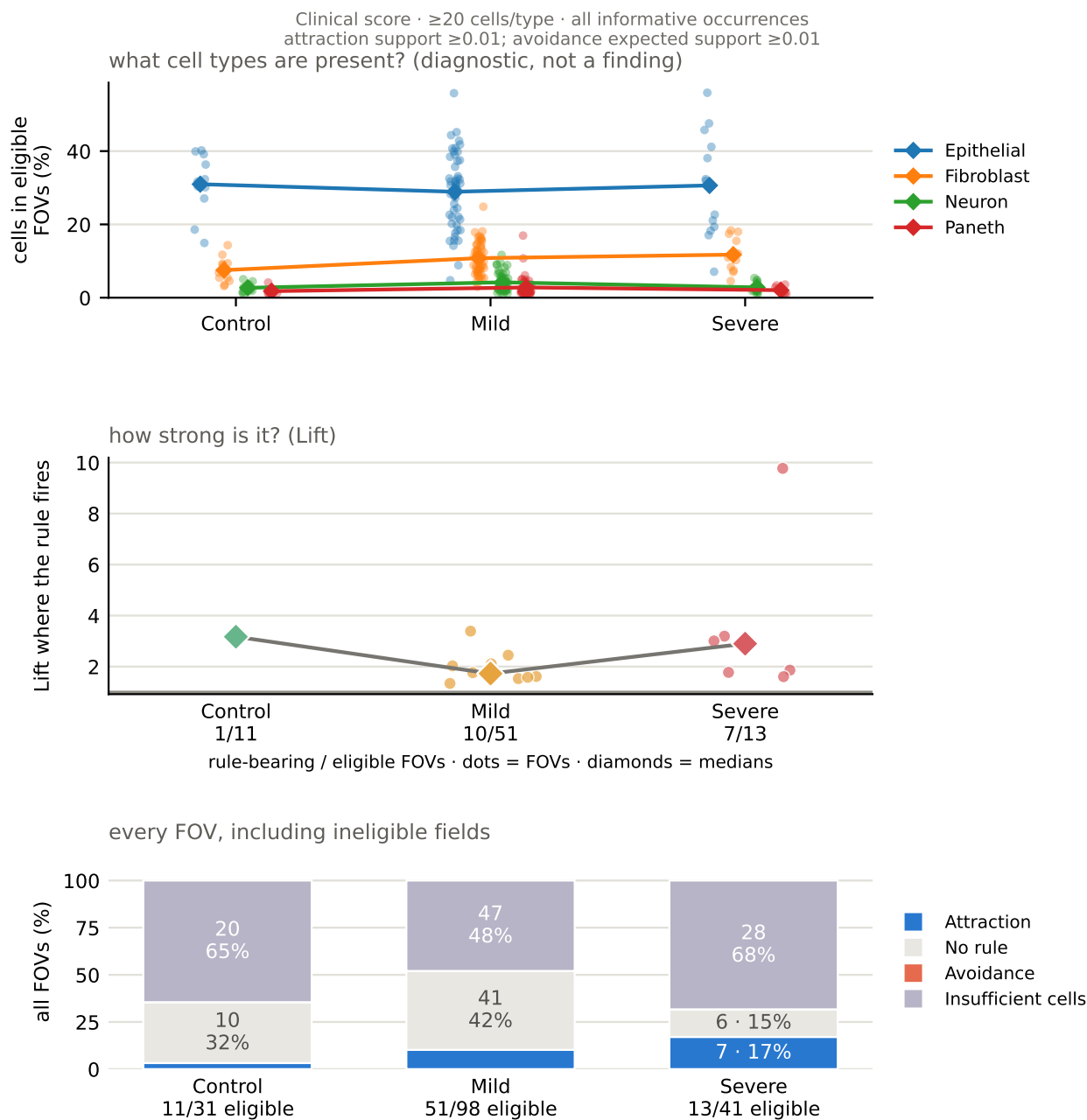


Figure 34: Fibroblast+Paneth → Epithelial+Neuron: 1/11 control, 10/51 mild, 7/13 severe eligible FOVs, 11 rule-bearing patients overall; patient-stage $q = 0.674$. This rule was selected by the baseline four-type screen.

At stricter support, almost all Paneth-related occurrences disappear.

Duodenum · Fibroblast + Paneth → Epithelial + Neuron

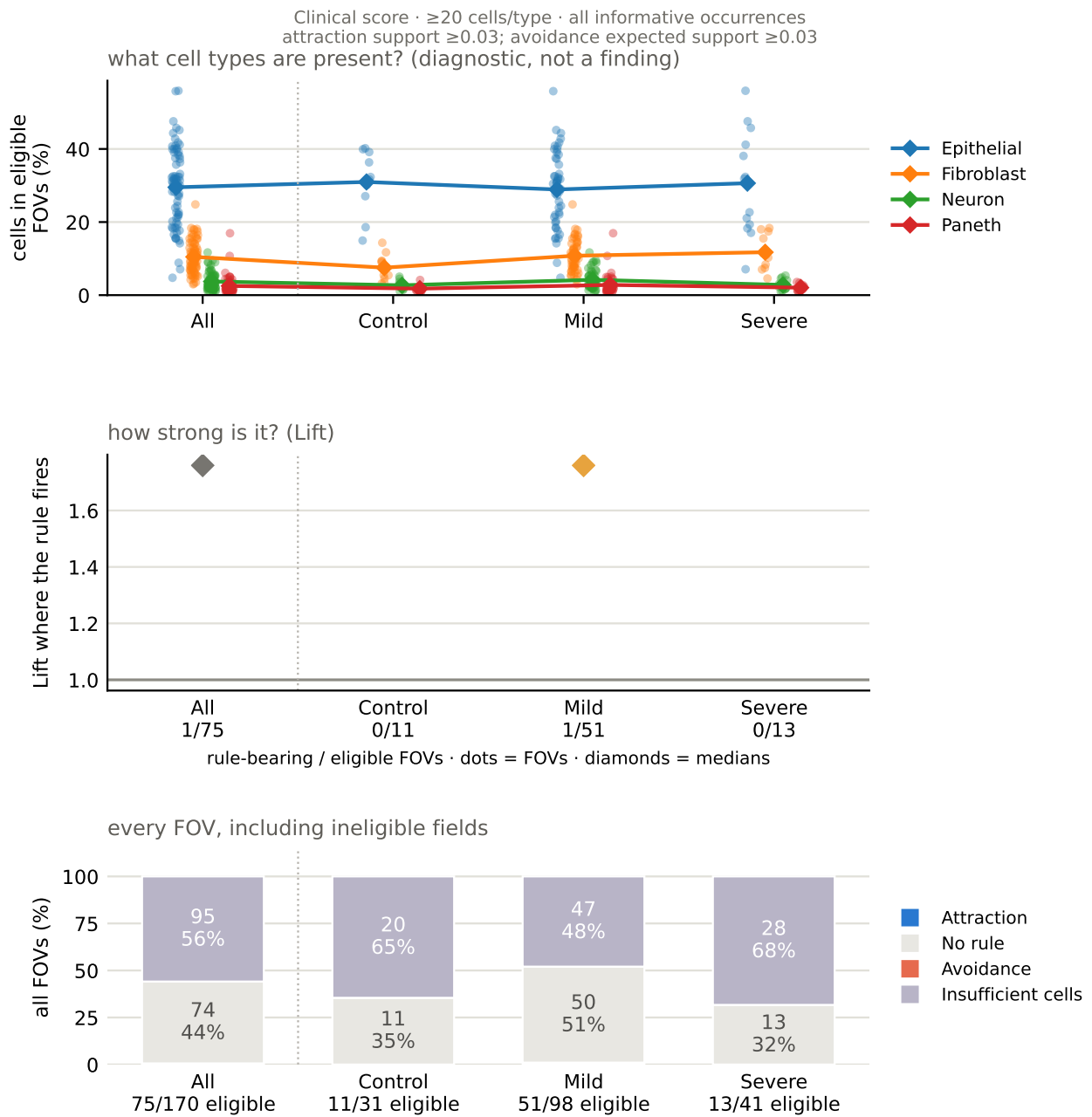


Figure 35: Optional check of the already selected rule: attraction support ≥ 0.03 leaves 1/75 rule-bearing eligible FOVs versus 18/75 at the baseline 0.01 gate. The mining output and eligibility denominator are unchanged; the strict gate is not a discovery filter.

3.5 Conditional strength and measured parent reversals

The Lift of a duodenal muscle–SMV attraction is stronger in mild than severe disease.

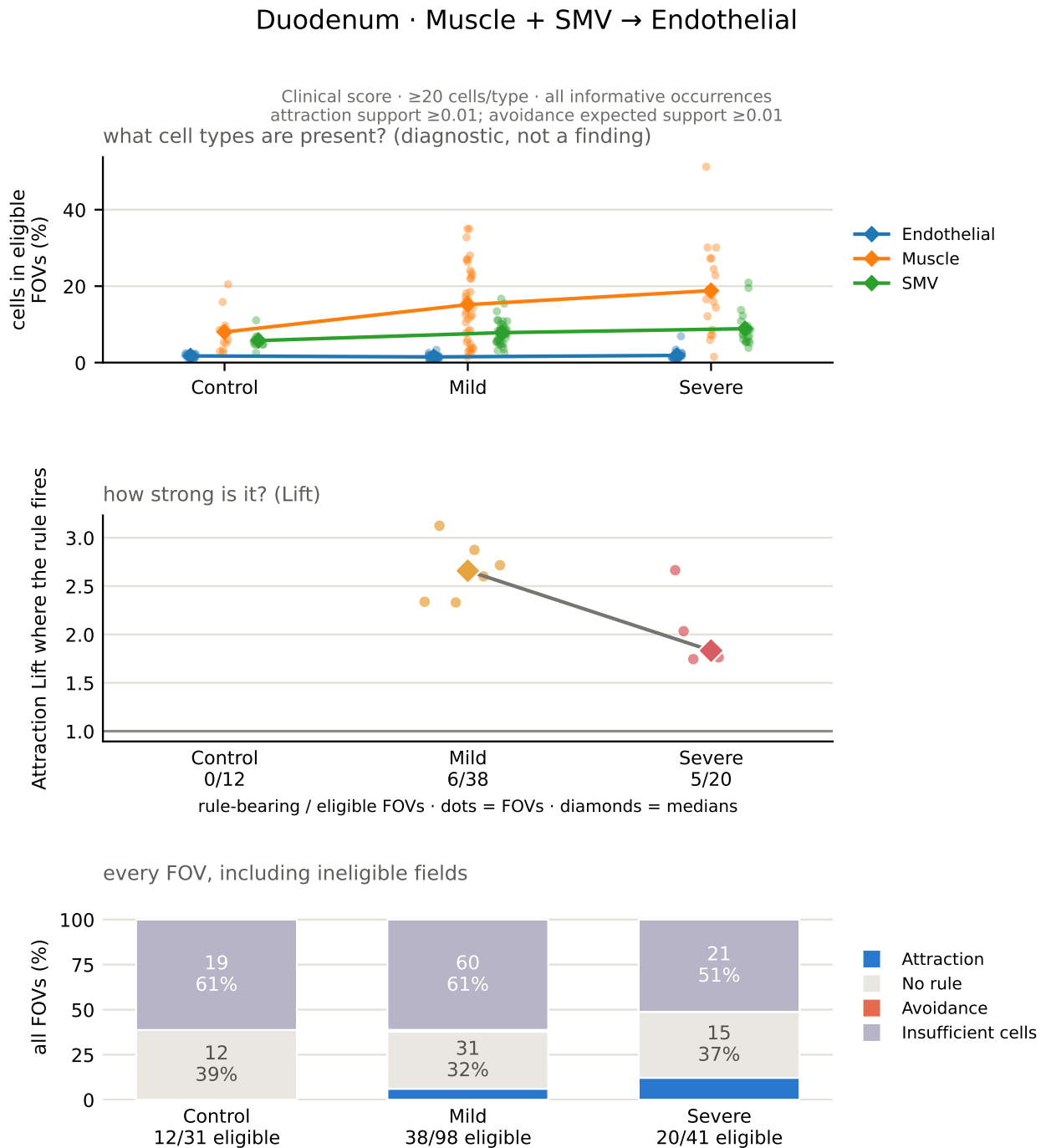


Figure 36: Muscle+SMV → Endothelial: 6/38 mild and 5/20 severe rule-bearing eligible FOVs, with median attraction Lift separated by 0.825; patient-level Lift $q = 0.651$. No control FOV bears this attraction. The strength screen compares only rule-bearing FOVs and never substitutes zero for absent rules.

Its tissue maps show the muscle and vessel arrangement behind the Lift contrast.

Duodenum · Muscle + SMV → Endothelial

Attraction examples · Mild

Unit: FOV · Score: clinical · Stages: Control / Mild / Severe · Eligibility: ≥ 20 cells/type

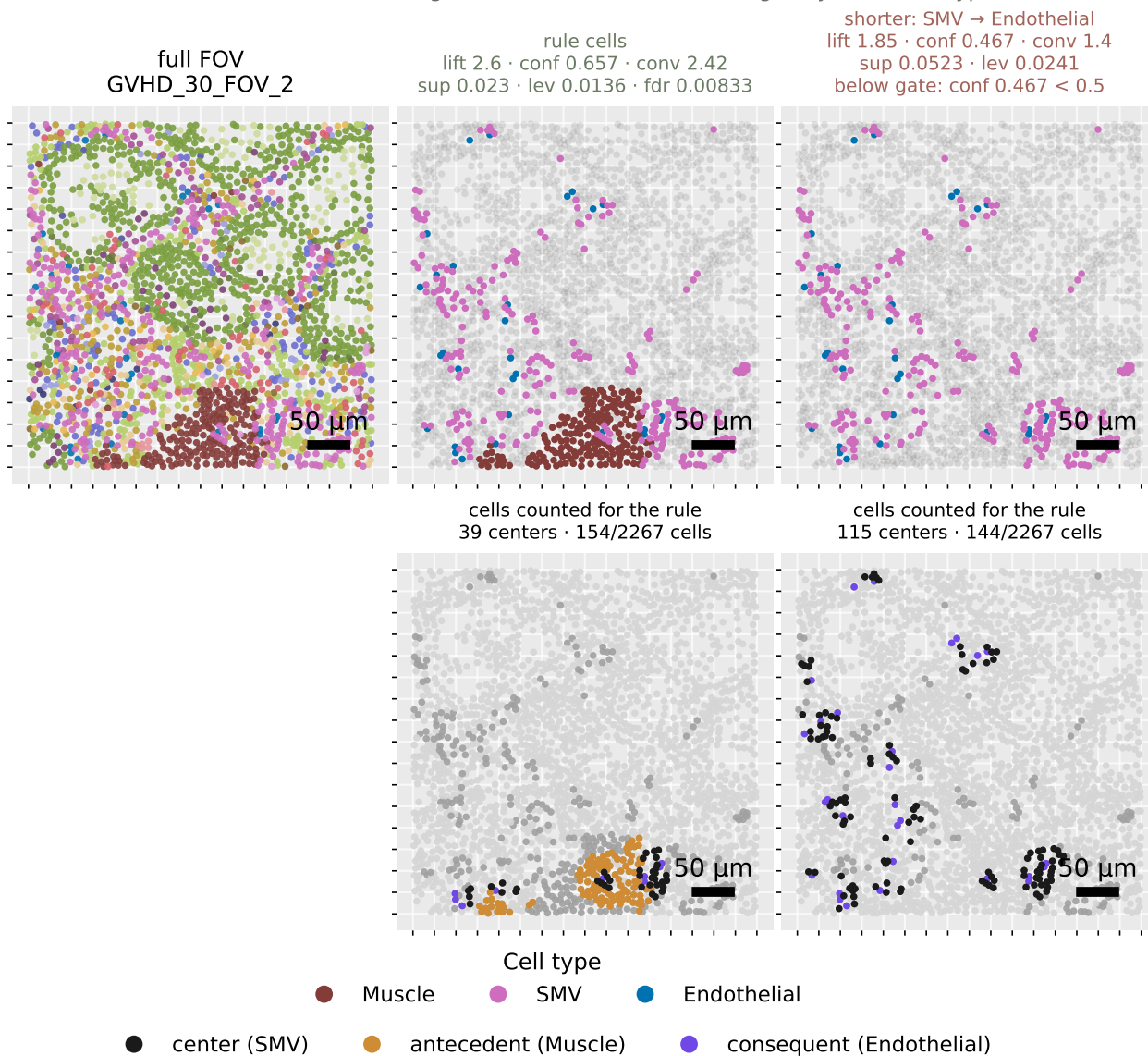


Figure 37: Representative mild maps as in Figure 4.

Adding fibroblast context reverses the stage trend of a duodenal epithelial-plasma rule.

Duodenum · Epithelial + Plasma → Fibroblast + Goblet

Complex rule versus its shorter parent

Parent: Epithelial + Plasma → Goblet · % of eligible FOVs

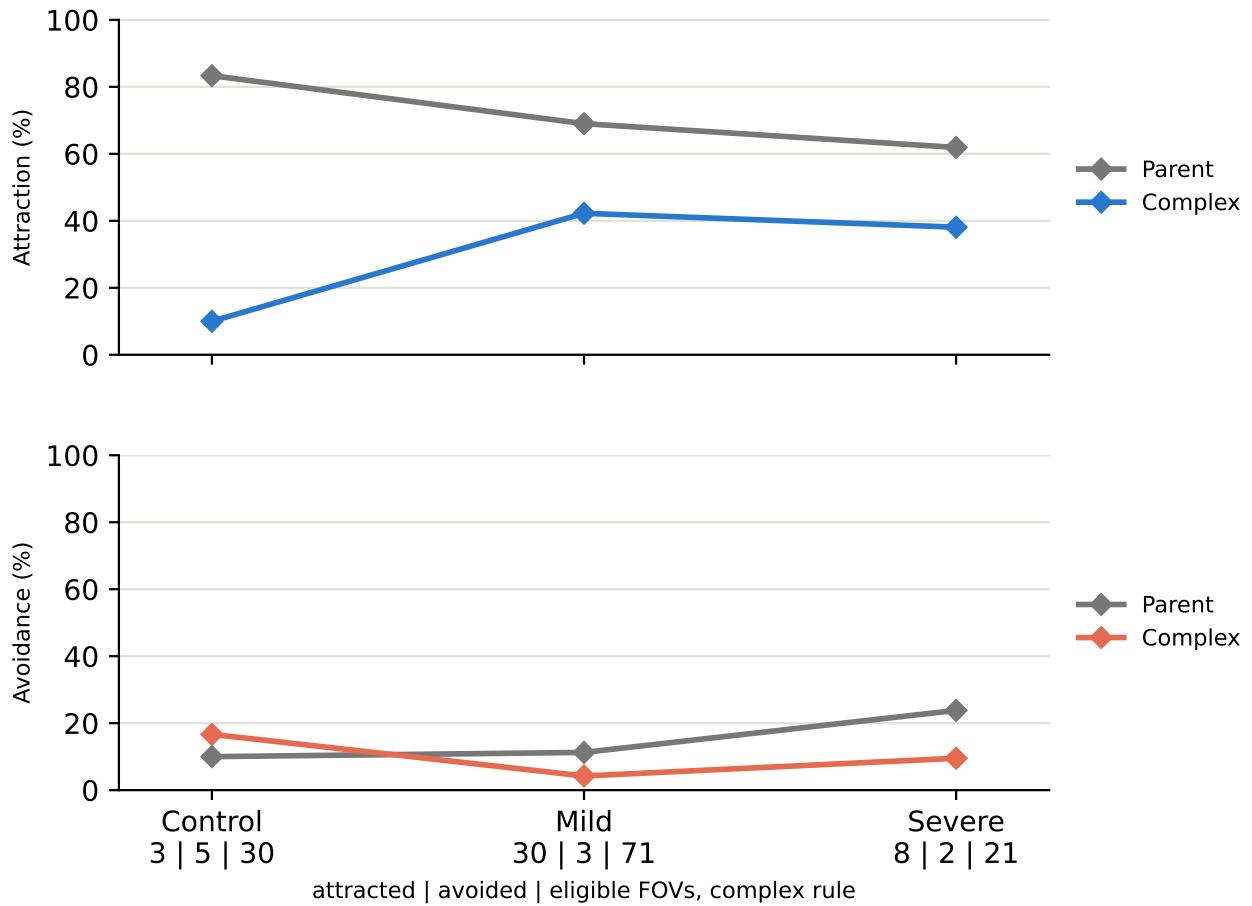


Figure 38: First reversal plot. Both measured rules use FOVs eligible for the complex rule; each point is attraction or avoidance FOVs divided by eligible FOVs for that stage. Parent Epithelial+Plasma → Goblet attraction is 25/30 control and 13/21 severe; complex Epithelial+Plasma → Fibroblast+Goblet is 3/30 and 8/21. The screen ranks opposing changes in these measured prevalences; 42 FOVs have matched passing parent and complex occurrences. It is descriptive, with no patient-level significance test.

A second duodenal parent-complex pair moves in opposite directions with severe disease.

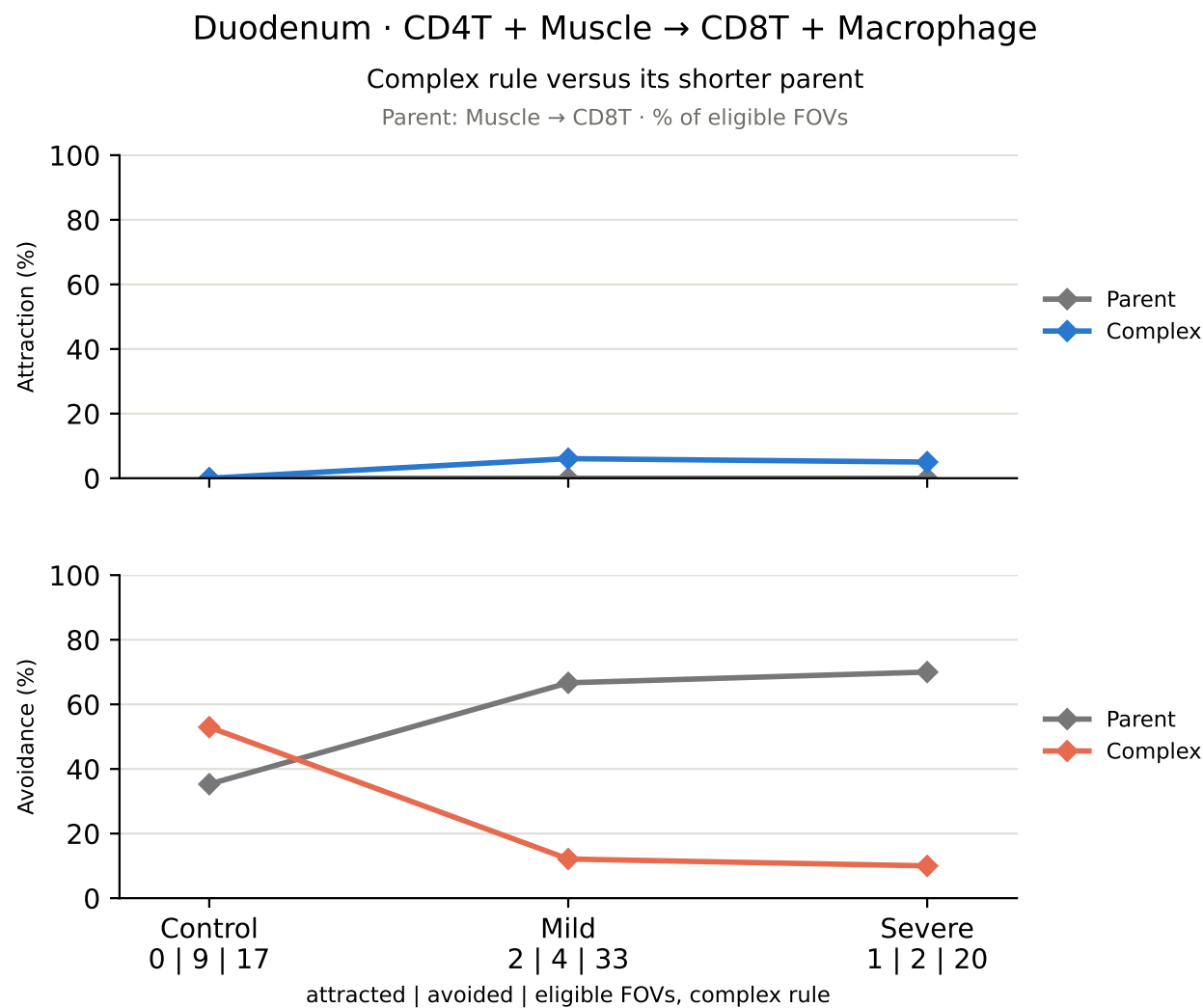


Figure 39: Parent Muscle → CD8T avoidance rises from 6/17 control to 14/20 severe eligible FOVs; complex CD4T+Muscle → CD8T+Macrophage avoidance falls from 9/17 to 2/20. Ten FOVs have matched passing occurrences. Calculation follows Figure 38.

The muscle-conditioned avoidance can be inspected in control tissue.

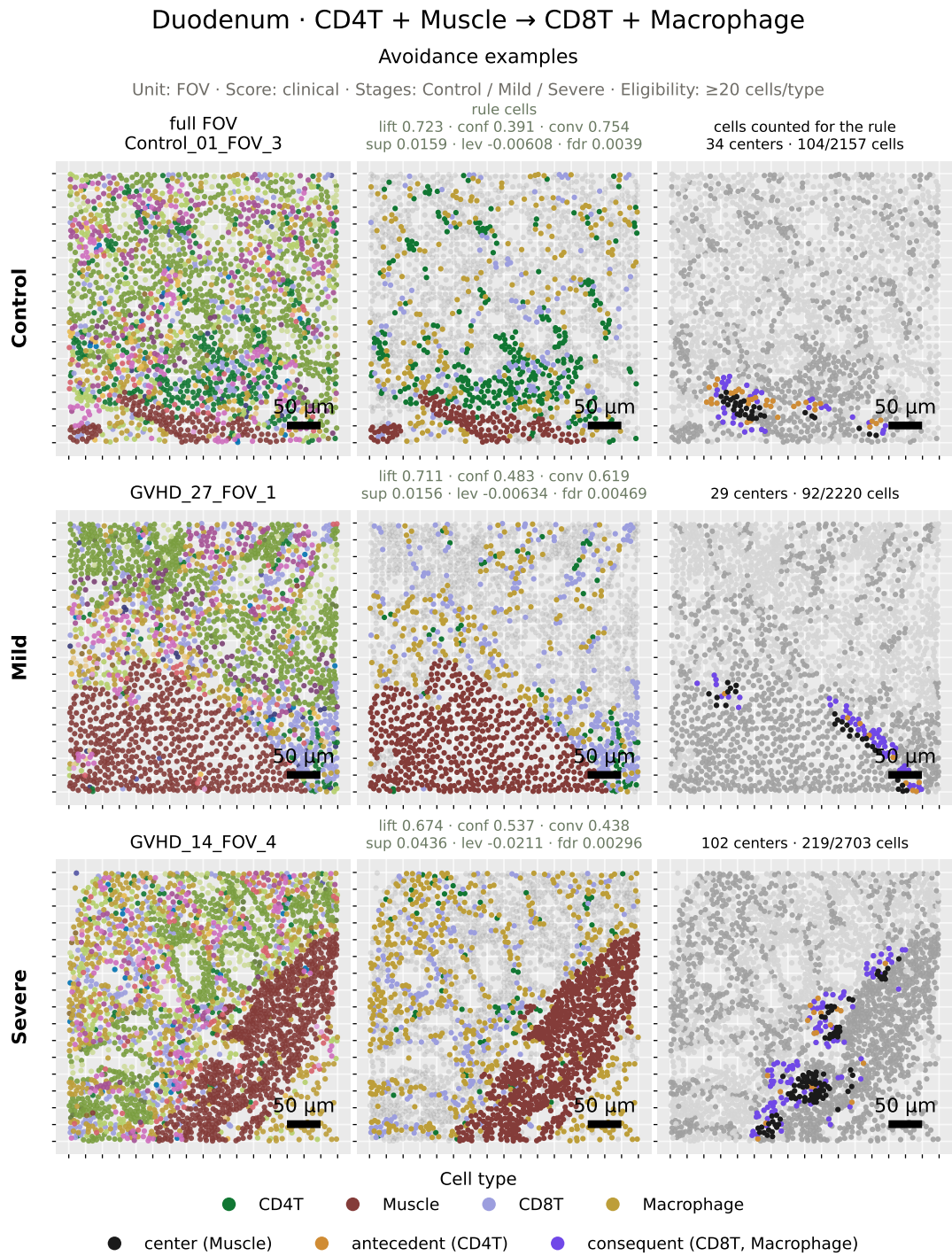


Figure 40: Representative map as in Figure 4.

A duodenal vascular parent rises while its CD4T-conditioned attraction vanishes in severe disease.

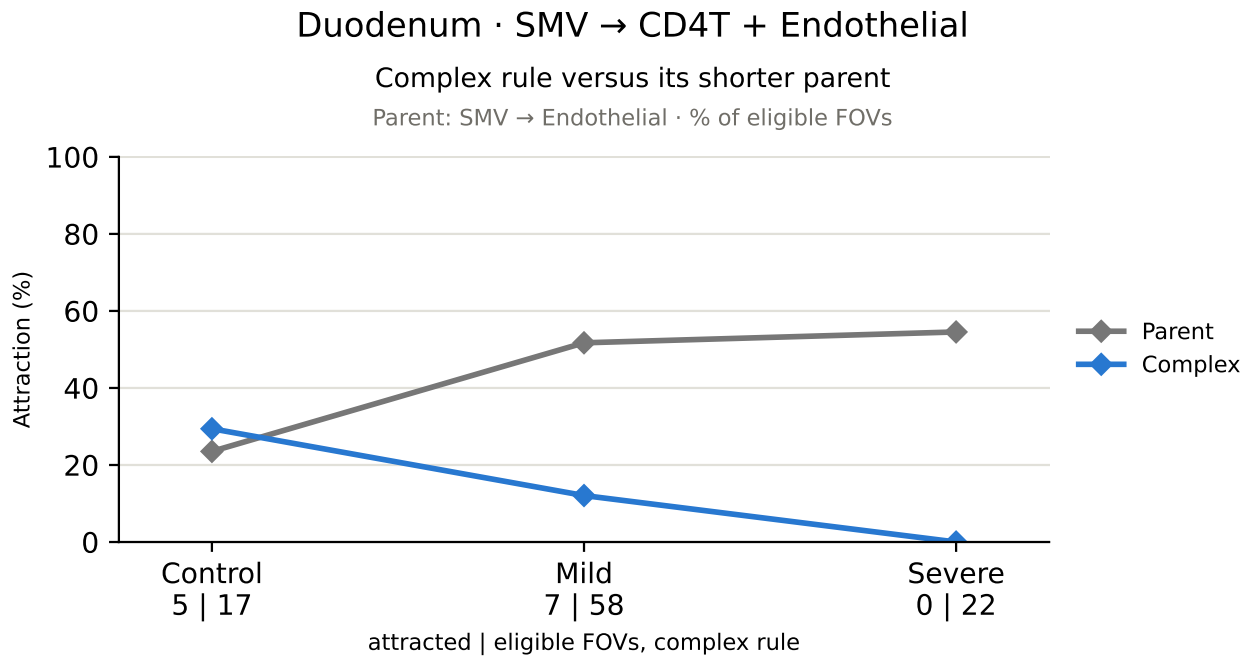


Figure 41: Parent SMV → Endothelial attraction: 4/17 control and 12/22 severe eligible FOVs. Complex SMV → CD4T+Endothelial attraction: 5/17 and 0/22; 15 matched passing FOVs. Descriptive comparison as in Figure 38.

A colon goblet attraction rises after adding SMV context even as its parent declines.

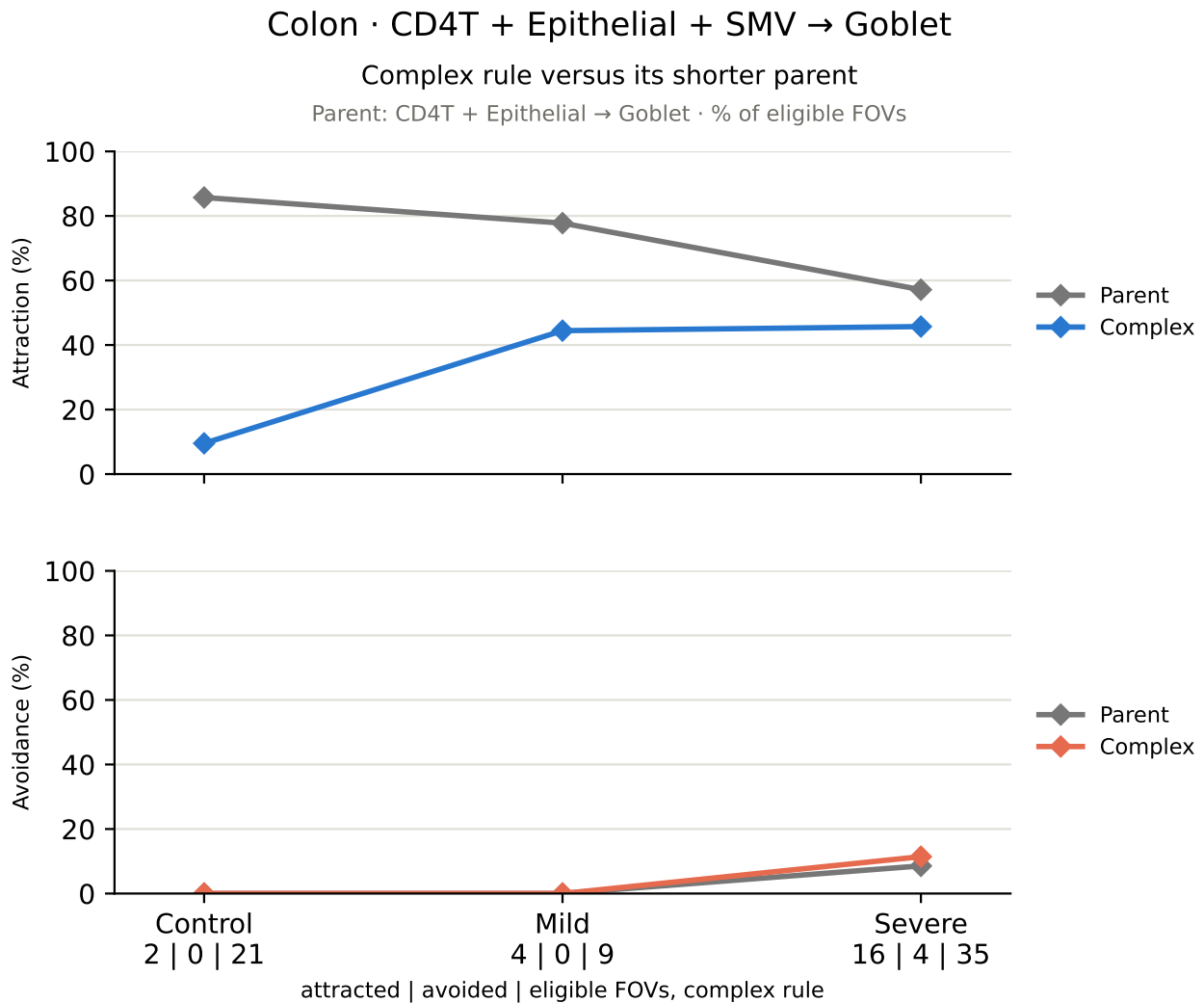


Figure 42: Parent CD4T+Epithelial → Goblet attraction: 18/21 control and 20/35 severe eligible FOVs. Complex CD4T+Epithelial+SMV → Goblet: 2/21 and 16/35; 23 matched passing FOVs. The two control complex FOVs are from one patient. Descriptive comparison as in Figure 38.